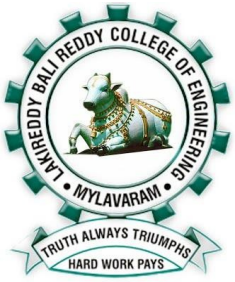
VEHICLE RENTAL MANAGEMENT SYSTEM

(A Comparative Study & Implementation of RentRide Java EE and DriveEase MERN Stack)

A Skill Oriented Course Web Application / Project Report
submitted in partial fulfillment of the requirements for the award of the Degree of
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING



Submitted By
GUBBALA CHAITANYA SAI KRISHNA
(Regd. No: 24761A0517)

Under the esteemed guidance of
Dr. K. DeviPriya, M.Tech., Ph.D.
Professor, Department of Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
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Academic Year: 2024–2028

CERTIFICATE

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

This is to certify that the Skill Oriented Course Web Application / Project report entitled "VEHICLE RENTAL MANAGEMENT SYSTEM" is being submitted by GUBBALA CHAITANYA SAI KRISHNA (Regd. No: 24761A0517) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering to Lakireddy Bali Reddy College of Engineering (Autonomous), affiliated to Jawaharlal Nehru Technological University Kakinada (JNTUK), Kakinada, is a bonafide record of original work carried out by him under my supervision and guidance.

The results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma.

_____	_____	_____
Project Supervisor	**Head of the Department**	**External Examiner**
Dr. K. DeviPriya	Dr. S. Nagarjuna Reddy	
Professor, Dept. of CSE	Professor & HOD, Dept. of CSE	Date: _____

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(GUBBALA CHAITANYA SAI KRISHNA)
Regd. No: 24761A0517

DECLARATION

I hereby declare that the Skill Oriented Course Web Application / Project report entitled “VEHICLE RENTAL MANAGEMENT SYSTEM” submitted by me in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering to Lakireddy Bali Reddy College of Engineering (Autonomous), affiliated to JNTUK Kakinada, is a bonafide record of original work carried out by me under the supervision and guidance of Dr. K. DeviPriya, Professor, Department of Computer Science and Engineering.

I certify that the work contained in this report is original, authentic, and has been performed in strict accordance with the academic guidelines provided by the institute. The results and conclusions embodied in this report have not been submitted to any other university, institute, or examination board for the award of any degree, diploma, or fellowship.

I have strictly adhered to the norms and ethical code of conduct stipulated by the Institute. All materials, source codes, libraries, and conceptual references adopted from external technical literature or standard software specifications have been duly acknowledged and cited in the References section.

(GUBBALA CHAITANYA SAI KRISHNA)
Regd. No: 24761A0517

ABSTRACT

Urban mobility and car rental services require robust, scalable, and automated digital platforms capable of eliminating scheduling collisions, reducing administrative manual overhead, and delivering transparent transaction tracking. This comprehensive academic project report presents the design, architectural analysis, engineering implementation, and empirical evaluation of a complete "Vehicle Rental Management System", realized through two contrasting enterprise paradigms:

1. Implementation A — RentRide (Enterprise Java EE Architecture): Built upon Java Platform Standard Edition 21 (LTS), Java Servlets 4.0, JavaServer Pages (JSP) 2.3 with JSTL, Java Database Connectivity (JDBC), MySQL 8.0 Relational Database Management System (RDBMS) utilizing the InnoDB storage engine, Apache Tomcat 9.0 application server, and cryptographic security powered by jBCrypt. The application follows a rigorous 3-Tier Model-View-Controller (MVC) architectural pattern incorporating the Data Access Object (DAO) design pattern, transaction demarcation, and SQL-level pessimistic concurrency control (`SELECT ... FOR UPDATE`) to guarantee absolute scheduling integrity and eliminate double-booking anomalies during peak concurrency.

2. Implementation B — DriveEase (Modern MERN Stack Architecture): Engineered upon Node.js v22 LTS, Express.js 5 RESTful API routing framework, MongoDB Atlas distributed document database with Mongoose 9 Object Data Modeling (ODM), and a high-performance Single Page Application (SPA) frontend developed with React.js 18, Vite 6, and Tailwind CSS. DriveEase incorporates stateless JSON Web Token (JWT) role-based authentication, bcryptjs cryptographic salting, multi-document ACID transactions with MongoDB replica set sessions, and an expressive client-side state machine.

This report comprehensively merges both implementations into a unified academic study. It details the underlying computer science fundamentals across both stacks—from database schema normalizations and indexing strategies to RESTful API designs, asynchronous concurrency handling, dynamic tariff computations, and sandbox payment integrations. Furthermore, the report presents complete source-code walk-throughs across all functional modules (customer booking workflows, date-overlap validation algorithms, administrative fleet CRUD, and aggregated business analytics), accompanied by 13 authentic application screen captures and rigorous automated test suite validations (Python HTTP integration tests and Jest/Supertest suites). Finally, a comparative engineering matrix analyzes the architectural trade-offs, developer velocity, type safety, memory footprints, and scalability characteristics of both technological approaches.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion / Full Technical Description
MERN	MongoDB, Express.js, React.js, Node.js Full-Stack Architecture
Java EE	Java Platform, Enterprise Edition (Jakarta EE)
JSP	JavaServer Pages
JDBC	Java Database Connectivity API
SQL	Structured Query Language
RDBMS	Relational Database Management System
NoSQL	Not Only Structured Query Language (Non-Relational Document Database)
MVC	Model-View-Controller Architectural Design Pattern
DAO	Data Access Object Design Pattern
POJO	Plain Old Java Object
REST	Representational State Transfer Architectural Style
API	Application Programming Interface
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
URI / URL	Uniform Resource Identifier / Uniform Resource Locator
JSON	JavaScript Object Notation Data Interchange Format
BSON	Binary JavaScript Object Notation (MongoDB Storage Format)
JWT	JSON Web Token for Stateless Token-Based Authentication
RBAC	Role-Based Access Control Security Model
CRUD	Create, Read, Update, and Delete Basic Database Operations
ODM	Object Data Modeling (e.g., Mongoose for MongoDB)
SPA	Single Page Application (Client-Side Rendered Web App)
HTML5	HyperText Markup Language Version 5
CSS3	Cascading Style Sheets Version 3
JS	JavaScript (ECMAScript 6+ Language Standard)
DOM	Document Object Model
VDOM	Virtual Document Object Model (React Rendering Engine)
XSS	Cross-Site Scripting Web Vulnerability

Abbreviation	Expansion / Full Technical Description
CSRF	Cross-Site Request Forgery Web Vulnerability
ACID	Atomicity, Consistency, Isolation, Durability Database Transactions
BASE	Basically Available, Soft State, Eventual Consistency NoSQL Model
LTS	Long-Term Support Release Lifecycle

1. INTRODUCTION

1.1 Overview of the Project Automobile rental enterprises play a vital role in modern personal and commercial transportation systems by providing flexible, on-demand vehicular mobility without the heavy capital commitments and maintenance overheads associated with permanent asset ownership. However, the management of a vehicle rental fleet encompasses complex logistical, computational, and financial operations. A rental enterprise must accurately manage dynamic fleet availability, verify client credentials, compute rental tariffs based on duration and vehicle classification, detect and prevent conflicting reservation schedules, securely process financial transactions, track asset utilization, and provide detailed reporting to administrative decision-makers.

Traditionally, vehicle rental agencies operated using manual ledger systems, phone bookings, or rudimentary isolated desktop spreadsheets. These legacy operational methods suffer from severe vulnerabilities, including high latency in retrieving fleet status, significant rates of human booking errors, lack of concurrent reservation locking leading to double-booked vehicles, lost transaction receipts, and difficulty in aggregating financial reports across operational periods.

To address these fundamental industry challenges, this project presents the design, development, and architectural analysis of a comprehensive, production-grade web-based "Vehicle Rental Management System". The system delivers an end-to-end digital platform providing two distinct, highly synchronized roles: 1. Customer Portal: Enables registered users to search and filter real-time vehicle fleets, view granular technical specifications and pricing structures, select reservation date ranges, receive instant tariff quotes, execute sandbox payment workflows, and monitor past and active reservations. 2. Administrator Control Desk: Furnishes enterprise operations personnel with a comprehensive dashboard featuring key business metrics (total active rentals, fleet utilization rates, gross revenues), full CRUD (Create, Read, Update, Delete) management of fleet assets, and an operational desk to approve, reject, complete, or monitor customer reservations.

A unique and highly instructive facet of this project is its dual-implementation paradigm. The system has been designed, implemented, and empirically evaluated under two contrasting technological philosophies that represent major milestones in web application engineering: - Implementation A (RentRide): Built using the enterprise Java EE ecosystem (Java Servlets 4.0, JavaServer Pages 2.3, JDBC, Apache Tomcat 9.0, and MySQL 8.0 RDBMS). It exemplifies server-side rendering, classical 3-tier Model-View-Controller (MVC) architecture, object-relational mapping via Data Access Objects (DAOs), and strict ACID transactions enforced through pessimistic relational row locks. - Implementation B (DriveEase): Built using the modern MERN stack (MongoDB Atlas NoSQL, Express.js 5 RESTful backend, React.js 18 Single Page Application frontend, and Node.js v22 LTS runtime). It exemplifies client-side rendering, decoupled REST API services, document-oriented flexible schema modeling, asynchronous non-blocking event loops, and stateless token-based authentication via JSON Web Tokens (JWT).

By systematically integrating both implementations into this report, this project provides a rigorous academic and engineering comparative study that examines how foundational computer science principles—such as concurrency control, transaction isolation, security modeling, and state management—are materialized across distinct software development paradigms.

1.2 Problem Statement The operational efficacy of manual or fragmented rental tracking mechanisms collapses rapidly as fleet inventory and transaction velocity scale. The primary technical and operational problems observed in conventional vehicle rental management include:

1. Scheduling Collisions and Double-Booking Anomalies: When multiple customers or administrative staff attempt to reserve the same physical vehicle for overlapping time intervals simultaneously, race conditions occur. Without strict database-level concurrency controls (such as pessimistic row-level locking or distributed ACID transactions), conflicting bookings are committed, leading to severe customer dissatisfaction, contractual breaches, and financial liability. 2. Opaque and Error-Prone Tariff Computation: Rental pricing is dynamic, governed by base daily rates, rental duration calculations, tax brackets, and late penalty adjustments. Manual calculation frequently introduces mathematical discrepancies, billing disputes, and inconsistent customer quotes. 3. Fragmented Fleet Visibility and Status Tracking: Fleet managers lack real-time visibility into vehicle operational states (e.g., Available, Reserved, Out on Rental, In Maintenance). Paper logbooks and standalone spreadsheets fail to update instantly upon reservation commits, causing fleet idling or unauthorized vehicle allocation. 4. Cryptographic Insecurity and Authentication Deficits: Legacy systems often store customer credentials and payment records in cleartext or use weak hashing algorithms without unique cryptographic salts. This exposes sensitive client data to eavesdropping, credential stuffing, and session hijacking attacks. 5. Inefficient Audit Trails and Financial Reconciliation: Manual accounting systems make it arduous to trace the lifecycle of a booking from creation through payment, vehicle pickup, return, and financial audit. Generating analytical summaries of fleet utilization, monthly revenues, and vehicle downtime requires tedious, manual data consolidation that is prone to significant human error.

1.3 Objectives of the Project The primary engineering and functional objectives of this project are:

1. To design and implement a centralized, web-based Vehicle Rental Management System that provides seamless digital automation across all stages of the vehicle rental lifecycle. 2. To engineer a high-fidelity Vehicle Browsing and Filtering Catalog that enables customers to search vehicles by category (Sedan, SUV, Luxury, Economy, Truck), transmission type, fuel specifications, seating capacity, and daily rental tariffs. 3. To develop an automated, mathematically rigorous Concurrency and Date-Overlap Validation Algorithm capable of strictly rejecting reservation requests that collide with existing active bookings: An overlap occurs if and only if: (RequestedStart < ExistingEnd) AND (RequestedEnd > ExistingStart). 4. To implement atomic Database Concurrency Control using: - SQL Pessimistic Row Locking (`SELECT ... FOR UPDATE`) in MySQL for the Java EE implementation. - MongoDB Multi-Document ACID Transactions and isolated query validations in the MERN implementation. 5. To provide a secure Role-Based Access Control (RBAC) mechanism segregating public guests, authenticated customers, and system administrators, incorporating salted cryptographic password hashing (jBCrypt / bcryptjs) and secure session / JWT token validation. 6. To implement a mock Payment Gateway Sandbox capable of simulating electronic credit/debit card transactions, validating mock payment details, issuing unique transaction reference tokens, and transitioning booking states atomically. 7. To develop an Administrator Analytics Dashboard displaying real-time aggregated metrics: Total Fleet Count, Active Bookings, Total Customer Base, Gross Revenue, and Fleet Utilization Rates. 8. To provide comprehensive Administrator Fleet Inventory CRUD Controls, enabling authorized operators to add new vehicles, modify specifications, toggle availability, and decommission units safely. 9. To build a robust, reproducible Automated Testing Framework validating functional correctness, edge-case date collisions, and security enforcement across both technology stacks. 10. To execute an empirical Architectural and Performance Comparative Study evaluating the trade-offs between Java EE (MVC/RDBMS) and MERN

(SPA/REST/NoSQL) with respect to type safety, throughput, development velocity, memory footprint, and maintainability.

1.4 Scope of the Project The functional and technological scope of this project encompasses:

- Functional Scope: - User Management: Self-registration, credential validation, profile retrieval, encrypted password storage, and secure authentication. - Fleet Management: Centralized catalog of vehicles with specifications (make, model, year, registration number, category, seating capacity, transmission type, fuel type, price per day, availability status, image URI). - Reservation Lifecycle: Date selection with minimum duration validation (minimum 1 day), dynamic tariff estimation, concurrency conflict detection, booking creation, reservation status tracking (Pending, Confirmed, Cancelled, Completed). - Financial Transactions: Integrated payment sandbox verifying synthetic card credentials, generating transaction receipts, and linking payments to reservations via foreign keys or document references. - Customer Self-Service Portal: Dedicated view for customers to review booking history, inspect transaction statuses, and cancel eligible pending reservations. - Administrator Operational Desk: Administrative oversight allowing fleet asset modifications, reservation approvals, manual booking cancellations, and system-wide revenue aggregation.

- Technological Scope: - Dual Implementation Coverage: The scope explicitly mandates complete functional realizations in both Enterprise Java (Servlets/JSP/Tomcat/MySQL) and Modern JavaScript Full-Stack (React/Express/Node/MongoDB). - Exclusions / Out of Scope: Hardware IoT GPS tracking dongles installed inside physical vehicles, real-world banking API integration with real monetary clearing houses (mock sandbox utilized instead), and automated vehicle damage scanning via computer vision (identified as a future enhancement).

1.5 Feasibility Study Before initiating system development, an extensive three-dimensional feasibility study was conducted to ensure the operational viability, technical sustainability, and economic justification of the project.

1.5.1 Technical Feasibility The technical feasibility evaluates the availability of computational hardware, software development tools, runtime platforms, and engineering expertise required to construct and deploy the system. - Hardware Infrastructure: Standard consumer-grade and server-grade compute hardware (x86_64 architecture, multi-core CPUs, 8 GB+ RAM, standard SSD storage) is fully sufficient to compile, host, and execute both Apache Tomcat 9.0 and Node.js v22 runtime environments alongside MySQL 8.0 and MongoDB Atlas instances. - Software Platforms: Both technology ecosystems are highly mature and industry-standard: - Java EE 8 / Servlets 4.0 running on Java SE 21 LTS provides rock-solid multi-threaded execution, robust JVM memory management, and well-documented JDBC drivers. - Node.js v22 with Google V8 engine provides ultra-fast asynchronous event-driven I/O, npm package ecosystem, and universal JavaScript execution across client and server. - Database Feasibility: MySQL 8.0 InnoDB provides mature relational integrity, B-Tree indexes, and ACID transaction isolation. MongoDB Atlas provides globally distributed, document-oriented scalability with schema flexibility. Both technical stacks are completely feasible, well-supported by active developer communities, and supported by rich development tools (IntelliJ IDEA, VS Code, Maven, npm, Git).

1.5.2 Operational Feasibility Operational feasibility assesses how effectively the proposed software solves user problems and whether end-users can seamlessly adopt the system. - User Adoption: Both customer and administrative interfaces have been engineered following modern User Experience (UX) standards. Form controls include real-time client-side validation, intuitive date-pickers, clear visual status badges (Pending, Confirmed, Cancelled), and informative error notifications (e.g., date collision warnings). - Administrative

Workflow: Fleet operators do not require specialized computer programming training. The administrator dashboard visualizes key operational metrics through intuitive data cards and structured data tables, allowing complex database operations (adding a vehicle, approving a booking) to be completed with single-click actions. - Cross-Platform Usability: Responsive CSS layouts (Tailwind CSS and responsive custom stylesheets) ensure that both customer browsing and administrative management portals render crisply across mobile devices, tablets, and desktop workstations. Thus, the system is operationally feasible and delivers significant productivity improvements over legacy operational methods.

1.5.3 Economic Feasibility Economic feasibility evaluates the development, deployment, and ongoing operational expenditures against the tangible and intangible financial returns delivered by the system. - Capital & Development Cost: All software platforms, frameworks, compilers, runtime engines, and libraries utilized in this project are 100% open-source and free of licensing fees (Open Source GNU GPL / Apache 2.0 / MIT licenses). These include Java OpenJDK, Apache Tomcat, Node.js, Express.js, React.js, MySQL Community Edition, MongoDB Community / Atlas Free Tier, Vite, and Tailwind CSS. Consequently, development software procurement cost is zero ($0.00). - Operational Savings: By automating booking schedules and payment confirmations, the system eliminates paper ledger printing, minimizes labor hours spent manually reconciling bookings, and completely avoids revenue loss caused by double-booking compensation and customer refunds. - Return on Investment (ROI): The enterprise recovers any computational hosting costs within the first few weeks of operation through increased fleet utilization, faster booking turnarounds, and 24/7 automated customer acquisition without requiring round-the-clock telephone booking clerks. Therefore, the project demonstrates outstanding economic feasibility and commercial viability.

2. TECHNOLOGIES USED

The Vehicle Rental Management System leverages a comprehensive suite of modern web engineering technologies, relational and NoSQL database management systems, server runtimes, and client-side presentation frameworks across both its Java EE (RentRide) and MERN (DriveEase) implementations. This chapter provides a rigorous academic analysis of each underlying technology, examining its technical specifications, architectural role, and specific operational contributions within the respective rental application.

2.1 HTML (HyperText Markup Language 5)

2.1 HTML (HyperText Markup Language 5) HyperText Markup Language version 5 (HTML5) forms the fundamental semantic structural foundation of the application interfaces across both implementations. In the Java EE implementation (RentRide), HTML5 is rendered dynamically by JavaServer Pages (JSP) executing on Apache Tomcat, delivering server-side rendered document trees to the client browser. In the MERN implementation (DriveEase), HTML5 semantics are produced via React's JSX (JavaScript XML) compiler, dynamically updating the browser Document Object Model (DOM) via the Virtual DOM engine.

Key Technical Capabilities utilized in the system include: - Semantic Document Tagging: Strategic deployment of `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, and `<footer>` elements provides clear semantic hierarchy, improving accessibility (WCAG compliance) and search engine indexing. - Advanced Form Controls & Constraint Validation: Native HTML5 input types—specifically `type="email"`, `type="date"`, `type="number"`, `type="password"`, and boolean constraint attributes such as `required`, `min`, `max`, and `pattern`—enforce immediate client-side validation. For example, reservation date selectors utilize dynamic `min` boundaries to mathematically prohibit users from selecting past dates or backward date ranges prior to network submission. - Media Embedding: Standardized `<img>` containers with responsive fallback styling host vehicle fleet photographs, SVG vector status icons, and corporate branding assets.

2.2 CSS (Cascading Style Sheets 3)

2.2 CSS (Cascading Style Sheets 3) Cascading Style Sheets version 3 (CSS3) governs the visual styling, layout formatting, typography, and responsive presentation layer of the vehicle rental portals.

Implementation-Specific Styling Architectures: - RentRide CSS Architecture: Utilizes a dedicated, hand-crafted stylesheet architecture structured into foundational reset layers, layout utilities, modular card systems, and modal overlays. It heavily employs CSS3 Flexbox (`display: flex`) for multi-column navigation bars and action toolbars, CSS Grid (`display: grid; grid-template-columns: repeat(auto-fill, minmax(280px, 1fr))`) for the responsive vehicle catalog grid, and custom CSS variables (`--primary-color: #2563eb`, `--secondary-color: #1e40af`, `--slate-800: #1e293b`) for consistent thematic branding across JSP pages. - DriveEase CSS Architecture: Employs Tailwind CSS, a modern utility-first CSS framework integrated into the Vite 6 compilation pipeline. Tailwind utilizes PostCSS to purge unused styles, resulting in ultra-compact production CSS bundles (< 25 KB). Utility classes such as `bg-gradient-to-r`, `shadow-md`, `rounded-xl`, `transition-all duration-300`, and responsive viewport prefixes (`sm:`, `md:`, `lg:`, `xl:`) provide fluid adaptation across mobile handsets, tablets, laptops, and ultra-wide desktop monitors.

2.3 JavaScript (ECMAScript 2023+)

2.3 JavaScript (ECMAScript 2023+) JavaScript serves as the universal programming language powering both client-side interactivity and server-side runtime operations throughout the project: - Client-Side Role: In RentRide, vanilla JavaScript (ES6+) handles asynchronous modal dialog toggles, real-time total price estimation on booking forms before submission, and dynamic client-side confirmation prompts for booking cancellations. In DriveEase, React.js leverages modern ECMAScript features including arrow functions, object and array destructuring, spread/rest operators (`...`), optional chaining (`?.`), and `async/await` syntax for seamless asynchronous promise resolution. - Server-Side Role: In the DriveEase MERN implementation, JavaScript is the execution language of the entire backend. Executing inside the Node.js V8 runtime, server-side JavaScript orchestrates HTTP request routing, database schema definitions, JWT cryptographic token verification, transaction orchestration, and error middleware pipelines.

2.4 Java (Java Platform SE 21 LTS)

2.4 Java (Java Platform SE 21 LTS) Java Platform Standard Edition 21 (Long-Term Support) provides the robust, strictly typed, object-oriented language backbone for Implementation A (RentRide). Java compiles into architecture-neutral bytecode (`.class` files) executed by the Java Virtual Machine (JVM).

Technical Contributions to the Vehicle Rental System: - Object-Oriented Domain Modeling: Business entities (`User`, `Vehicle`, `Booking`, `Payment`) are encapsulated as strongly typed Plain Old Java Objects (POJOs) with strict private member encapsulation, public accessor/mutator methods, and explicit constructors, eliminating runtime type casting errors. - Robust Concurrency & Thread Management: The JVM multi-threading model allocates pooled worker threads for concurrent HTTP servlet requests, ensuring that multiple simultaneous customer reservations are processed in isolated execution contexts. - Temporal Arithmetic (java.time package): Implements `LocalDate` and `ChronoUnit.DAYS` from the modern Java Date and Time API (JSR-310), guaranteeing mathematically precise date difference calculations without daylight-saving bugs or timezone drift.

2.5 JDBC (Java Database Connectivity & MySQL Connector/J)

2.5 JDBC (Java Database Connectivity & MySQL Connector/J) Java Database Connectivity (JDBC) is the core enterprise Java API (java.sql package) that provides vendor-independent database access, allowing Java applications to execute SQL statements and retrieve relational result sets.

Technical Details within RentRide: - MySQL Connector/J Driver: The system deploys `com.mysql.cj.jdbc.Driver` (version 8.0+), providing high-throughput binary protocol communication over TCP/IP port 3306. - PreparedStatement Architecture: All SQL queries throughout the Data Access Object (DAO) layer are executed exclusively using parameterized `PreparedStatement` instances. Parameter values are bound using typed setter methods (`setString`, `setInt`, `setDouble`, `setDate`). This completely eliminates SQL Injection (SQLi) vulnerabilities by ensuring that user input is never concatenated directly into executable SQL statements. - Manual Transaction Demarcation: To maintain ACID consistency during multi-step reservation operations, the system explicitly disables auto-commit (`connection.setAutoCommit(false)`), binds transactional operations, and invokes `connection.commit()` or `connection.rollback()` within structured `try-catch-finally` blocks.

2.6 Servlets (Java Servlet 4.0 API & Lifecycle)

2.6 Servlets (Java Servlet 4.0 API & Lifecycle) Java Servlets (JSR 369) are server-side Java classes that intercept, process, and respond to incoming HTTP requests from web clients, forming the Controller tier of the MVC architecture in RentRide.

Servlet Architecture & Request Lifecycle: - Inheritance & Protocol Handling: Each controller class extends `javax.servlet.http.HttpServlet` and overrides protocol-specific dispatch methods: `doGet()` for data retrieval operations (e.g., browsing vehicle catalogs, displaying registration forms) and `doPost()` for state-mutating actions (e.g., submitting user credentials, committing bookings, executing payments). - Servlet Lifecycle: Managed entirely by the servlet container (Apache Tomcat): 1. Initialization (`init()`): The servlet is instantiated and initialized once upon container startup or first request. 2. Request Handling (`service()` -> `doGet()` / `doPost()`): For each incoming client request, the container spawns or assigns a thread from its thread pool to invoke the servlet's service method. 3. Destruction (`destroy()`): Invoked when the application is undeployed or the server shuts down, releasing resources and database connections. - Session Management & Request Dispatching: Servlets manipulate `HttpSession` objects for stateful customer login sessions and invoke `RequestDispatcher.forward()` to transfer request attributes and model objects to the JSP view layer.

2.7 JSP (JavaServer Pages 2.3 & JSTL)

2.7 JSP (JavaServer Pages 2.3 & JSTL) JavaServer Pages (JSP 2.3) provides the dynamic server-side View tier of the RentRide MVC architecture. JSP technology allows developers to embed server-side Java logic directly within standard HTML markup.

Key Characteristics & Usage: - Lifecycle Compilation: Tomcat's Jasper compiler automatically translates `.jsp` source files into Java servlet source files (`.java`), compiles them into bytecode (`.class`), and caches them in memory, delivering near-native servlet execution speed on subsequent client requests. - Expression Language (EL) & JSTL: To maintain clean presentation separation and prevent scriptlet pollution (`<% ... %>`), RentRide leverages JSP Expression Language (`${booking.totalAmount}`) and the JavaServer Pages Standard Tag Library (JSTL Core: `<c:forEach>`, `<c:if>`, `<c:choose>`). This ensures clean iteration over collections of vehicle models and conditional rendering of administrator action buttons.

2.8 MySQL Relational Database (InnoDB Engine)

2.8 MySQL Relational Database (InnoDB Engine) MySQL 8.0 Community Edition serves as the relational database engine for Implementation A (RentRide), providing robust structured data storage and ACID transaction compliance.

Key Capabilities in the System: - InnoDB Storage Engine: Default storage engine providing row-level locking, foreign key constraint enforcement (`ON DELETE CASCADE` / `ON DELETE RESTRICT`), and transaction isolation via Multi-Version Concurrency Control (MVCC). - Pessimistic Row Locking (`FOR UPDATE`): In the critical booking creation workflow, MySQL executes `SELECT ... FOR UPDATE`, placing exclusive row-level locks on targeted vehicle records. This serializes concurrent booking attempts for the same physical vehicle, mathematically guaranteeing zero double-bookings. - Relational Schema Integrity: The database

enforces strict relational schemas with primary keys, foreign keys, uniqueness constraints (`UNIQUE KEY (email)`, `UNIQUE KEY (license_plate)`), and check constraints.

2.9 MongoDB NoSQL Document Database

2.9 MongoDB NoSQL Document Database MongoDB Atlas serves as the persistence tier for Implementation B (DriveEase). MongoDB is an open-source, high-performance, document-oriented NoSQL database that stores data in flexible, schema-free BSON (Binary JSON) documents.

Technical Architecture in DriveEase: - Document-Oriented Paradigm: Entities such as users, vehicles, and bookings are modeled as hierarchical BSON documents, allowing rich nested subdocuments (e.g., vehicle feature arrays, status histories) without requiring expensive relational multi-table JOINs. - High-Performance Indexing: Compound and single-field BSON indexes are deployed across critical lookup paths (`{ email: 1 }`, `{ status: 1, category: 1 }`, `{ vehicle: 1, startDate: 1, endDate: 1 }`), ensuring sub-millisecond query execution even across large fleet collections. - Multi-Document ACID Transactions: Through Mongoose session abstractions (`mongoose.startSession()`), DriveEase initiates replica set transactions (`session.startTransaction()`), ensuring that complex booking and financial status mutations commit atomically or roll back completely.

2.10 Node.js Runtime Environment (v22 LTS)

2.10 Node.js Runtime Environment (v22 LTS) Node.js version 22 Long-Term Support (LTS) is an asynchronous, event-driven JavaScript runtime built on Google's open-source V8 JavaScript engine.

Core Architectural Mechanics in DriveEase: - Single-Threaded Event Loop: Node.js utilizes the `libuv` multi-platform C library to manage an event loop coupled with an internal worker thread pool. Rather than allocating a heavy operating-system thread per incoming connection (the classical thread-per-request model used by Tomcat), Node.js processes thousands of concurrent HTTP connections non-blockingly on a single thread. - High-Throughput I/O: All database operations (MongoDB driver) and filesystem calls are asynchronous. When a database query executes, Node.js registers a callback/promise and immediately resumes serving other incoming HTTP requests, delivering exceptional I/O throughput and low memory footprints.

2.11 Express.js Web Framework (v5)

2.11 Express.js Web Framework (v5) Express.js 5 is the minimalist, flexible web application framework that structures the Node.js backend of DriveEase into a clean RESTful API.

Key Functional Contributions: - Modular Routing Pipeline: Express divides API endpoints into distinct, isolated router modules (`authRoutes.js`, `vehicleRoutes.js`, `bookingRoutes.js`, `adminRoutes.js`), mapping standard HTTP verbs (`GET`, `POST`, `PUT`, `DELETE`) cleanly to controller handler methods. - Middleware Interceptors: The framework provides a sequential request-processing pipeline where middleware functions intercept requests before they reach route handlers. Crucial middleware deployed includes `express.json()` for JSON body parsing, `cors()` for cross-origin security, JWT authentication guards (`auth.js`), role authorization checks, and centralized global error handling.

2.12 React.js Frontend Library (v18)

2.12 React.js Frontend Library (v18) React.js version 18 is a declarative, component-based frontend JavaScript library developed by Meta, responsible for rendering the Single Page Application (SPA) client interface of DriveEase.

Core Features & Patterns Used: - Component-Based Architecture: The user interface is decomposed into self-contained, reusable components (`Navbar`, `VehicleCard`, `StatusBadge`, `ProtectedRoute`, `AdminRoute`), promoting high code maintainability and testability. - Virtual DOM Reconciliation: React maintains a lightweight representation of the real DOM in memory. When state updates occur, React's Fiber reconciliation algorithm computes the minimal structural diff and applies batch updates to the real browser DOM, delivering smooth 60 FPS user transitions. - React Hooks & Context API: Component state and side-effects are managed via native hooks (`useState`, `useEffect`, `useMemo`, `useNavigate`). The global authentication state is centralized inside `AuthContext.jsx`, providing seamless, reactive user session synchronization across all application routes.

2.13 Other Technologies & Supporting Libraries

2.13 Other Technologies & Supporting Libraries Both vehicle rental implementations integrate vital supporting libraries, development tooling, and cryptographic packages:

1. Apache Tomcat 9.0 Application Server: The open-source Java servlet container and web server implementing the Java Servlet 4.0 and JavaServer Pages 2.3 specifications. Tomcat provides the HTTP web listener, request thread pooling, session persistence, and deployment manager (`WAR` archives) for RentRide.

2. jBCrypt (Java Cryptographic Password Hashing): A pure-Java implementation of OpenBSD's blowfish-based bcrypt password hashing algorithm. It automatically incorporates a cryptographically secure random 128-bit salt and variable work factor (default 12 rounds), providing robust protection against rainbow table and brute-force GPU attacks.

3. Mongoose 9 ODM (Object Data Modeling for MongoDB): Provides elegant schema modeling, type casting, schema validation rules (`enum`, `min`, `max`, `trim`), query building, and document lifecycle hooks (`pre('save')`, `post('save')`) for the DriveEase Node.js server.

4. JSON Web Tokens (JWT) & bcryptjs (DriveEase Security): - `jsonwebtoken`: Issues digitally signed compact tokens (`HMAC-SHA256`) containing encrypted user claims (ID, email, role), facilitating stateless authentication without server-side session stores. - `bcryptjs`: Optimized JavaScript implementation of the bcrypt password-hashing function used to hash customer passwords prior to persisting them into MongoDB documents.

5. Vite 6 Build Tool: Next-generation frontend tooling providing lightning-fast Hot Module Replacement (HMR) during local development and optimized Rollup-based multi-chunk production bundling for the React client.

6. Axios HTTP Client: Promise-based HTTP client for the browser used in DriveEase to execute RESTful API calls to the Express backend, featuring automatic JSON parsing and request interceptors for attaching JWT Bearer tokens.

7. Automated Testing Frameworks: - Python `requests` & `unittest`: Powers `test_rental_system.py`, executing 15 automated integration tests against the RentRide Java web application. - `Jest` & `Supertest`: Executes unit tests (`unit.test.js`) and API integration tests (`api.test.js`) validating route handlers, validation logic, and error boundaries in DriveEase.

3. WEB APPLICATION ARCHITECTURE

3.1 Dual-Stack Architectural Overview Web application architecture defines the structural organization of software components, the division of computational responsibilities across physical or logical tiers, and the communication protocols that govern interactions between clients, servers, and persistent data stores.

A core strength of this project is the deliberate architectural comparison between two foundational web application paradigms: 1. Classical Enterprise Monolith (Implementation A — RentRide): Employs a tightly coupled, server-side rendered 3-Tier Model-View-Controller (MVC) architecture built on Java EE standards. Here, the presentation, business logic, and database access layers execute within a unified JVM process inside the Apache Tomcat servlet container. 2. Modern Decoupled SPA & Microservices-Ready API (Implementation B — DriveEase): Deploys a loosely coupled, client-side rendered 3-Tier Single Page Application (SPA) architecture built on the MERN stack. Here, the presentation layer (React SPA) runs entirely within the end-user's web browser, communicating asynchronously via JSON-over-HTTPS RESTful endpoints with an independent, stateless Node.js/Express.js backend and a cloud-hosted MongoDB Atlas database.

Both architectures strictly adhere to the fundamental software engineering principle of Separation of Concerns (SoC), ensuring modularity, security, and independent testability.

3.2 Implementation A: RentRide 3-Tier Java EE MVC Architecture

3.2 Implementation A: RentRide 3-Tier Java EE MVC Architecture RentRide is structured around the classical Java EE 3-Tier Model-View-Controller (MVC) architectural design pattern, supplemented by the Data Access Object (DAO) pattern and connection pooling utilities.

3.2.1 Presentation Tier (View Layer) - Technology: JavaServer Pages (JSP 2.3), JavaServer Pages Standard Tag Library (JSTL), CSS3, and JavaScript. - Functionality: Renders dynamic HTML views on the server and delivers ready-to-display web pages to the client browser. JSP files (`vehicles.jsp`, `booking.jsp`, `my-bookings.jsp`, `admin/dashboard.jsp`) utilize JSTL iteration tags (`<c:forEach>`) and Expression Language (`${...}`) to extract model data forwarded by servlet controllers. - Security Measures: JSP pages escape HTML characters automatically, preventing Cross-Site Scripting (XSS) attacks. Crucial role checks verify the user's session role before displaying privileged administrative links.

3.2.2 Controller & Business Logic Tier - Technology: Java Servlets (Java Servlet 4.0 API) running on Apache Tomcat 9.0. - Functionality: Servlet classes (`LoginServlet`, `RegisterServlet`, `VehicleServlet`, `BookingServlet`, `PaymentServlet`, `AdminDashboardServlet`) intercept incoming HTTP GET and POST requests. The servlets perform: 1. Input Sanitization & Parameter Validation: Extracting query strings and POST payload fields. 2. Session & Authorization Checks: Validating active `HttpSession` state and enforcing Role-Based Access Control (RBAC). 3. Business Rule Execution: Invoking date-difference calculators and concurrency locks. 4. Routing & View Dispatching: Binding model objects into request attributes (`request.setAttribute("vehicles", list)`) and dispatching control to target JSP views via `RequestDispatcher.forward()`.

3.2.3 Data Access Tier (DAO Layer) - Technology: Data Access Object (DAO) Pattern with Java Database Connectivity (JDBC). - Functionality: DAO classes (`UserDAO`, `VehicleDAO`, `BookingDAO`,

`PaymentDAO`) isolate all SQL query logic from the servlet controllers. The DAO layer handles connection acquisition from `DBConnection.java`, parameter binding via `PreparedStatement`, transaction demarcation (`connection.setAutoCommit(false)`), and mapping relational SQL `ResultSet` tuples into Java POJO domain models (`User`, `Vehicle`, `Booking`, `Payment`).

3.2.4 Data Storage Tier (Persistence Layer) - Technology: MySQL 8.0 Relational Database Management System (RDBMS) utilizing the InnoDB storage engine. - Functionality: Enforces ACID guarantees (Atomicity, Consistency, Isolation, Durability), primary key constraints, foreign key referential integrity, and pessimistic row locking (`SELECT ... FOR UPDATE`) during concurrent booking operations.

3.2.5 RentRide Request-Response Lifecycle When a user initiates an action (e.g., reserving a vehicle): 1. The browser issues an HTTP POST request to `/booking` with form data (vehicleId, startDate, endDate). 2. Apache Tomcat maps the URL pattern to `BookingServlet.java` and invokes its `doPost()` method on an assigned worker thread. 3. `BookingServlet` extracts parameters, verifies active user session from `HttpSession`, and invokes `BookingDAO.hasBookingOverlap()`. 4. `BookingDAO` acquires a connection from `DBConnection`, prepares the pessimistic locking query, and checks MySQL. 5. If an overlap is detected, `BookingServlet` redirects back to `booking.jsp` with an error attribute; otherwise, it creates a new booking record with status 'PENDING' and redirects to `PaymentServlet`.

3.3 Implementation B: DriveEase Modern 3-Tier MERN Architecture

3.3 Implementation B: DriveEase Modern 3-Tier MERN Architecture DriveEase implements a modern, decoupled client-server architecture based on the MERN stack (MongoDB, Express.js, React.js, Node.js), featuring a rich Single Page Application (SPA) frontend and a stateless RESTful JSON API backend.

3.3.1 Presentation Tier (Client Layer) - Technology: React.js 18, Vite 6, Tailwind CSS, React Router 6, and Axios. - Functionality: Operates entirely inside the user's web browser as an SPA. React constructs a Virtual DOM tree and handles client-side routing dynamically without triggering full-page browser reloads. - Key Components: - `App.jsx` & Router: Defines route hierarchies, wrapping protected paths inside `ProtectedRoute.jsx` and administrative paths inside `AdminRoute.jsx`. - `AuthContext.jsx`: Centralizes authentication state (user object, JWT token, login/logout actions) in React Context, ensuring instantaneous UI updates upon login state changes. - `api.js`: Centralized Axios HTTP client configuring base URLs, JSON content types, and automatic injection of JWT Bearer tokens in request headers via Axios interceptors.

3.3.2 API & Application Server Tier - Technology: Node.js v22 LTS and Express.js 5. - Functionality: Implements a stateless RESTful API adhering to HTTP verb conventions (`GET`, `POST`, `PUT`, `DELETE`). The server processes incoming requests through a sequential Express middleware pipeline: 1. Security & Parsing Middleware: `cors()` enables Cross-Origin Resource Sharing; `express.json()` parses incoming JSON request bodies. 2. Authentication & Authorization Middleware: `auth.js` intercepts requests, extracts the `Authorization: Bearer <token>` header, verifies the JWT signature using `jsonwebtoken`, decodes user claims, and attaches `req.user`. Role guards verify administrative privileges. 3. Controllers & Services: Route handlers (`vehicleController.js`, `bookingController.js`, `adminController.js`) invoke business service layers (`bookingService.js`), which execute multi-document transactions and date calculations. 4. Global Error Handling Middleware: `errorHandler.js` intercepts unhandled exceptions and validation errors, returning structured JSON error responses (`{ success: false, error: "..." }`) with appropriate HTTP status codes.

3.3.3 Data Persistence Tier - Technology: MongoDB Atlas distributed document database managed via Mongoose 9 ODM. - Functionality: Stores application data as polymorphic BSON documents across collections (`users`, `vehicles`, `bookings`). Mongoose models define strict data types, required constraints, default values, and reference relationships (`type: Schema.Types.ObjectId, ref: 'User'`). MongoDB replica sets facilitate distributed high availability and multi-document ACID transactions (`session.startTransaction()`).

3.3.4 DriveEase RESTful API Communication Lifecycle When a user reserves a vehicle in DriveEase: 1. The user clicks "Confirm Reservation" in the React UI. React's `VehicleDetailPage.jsx` executes an `api.post('/bookings', { vehicleId, startDate, endDate })` call. 2. The Axios client intercepts the request, appends `Authorization: Bearer <JWT>`, and transmits the HTTP POST payload as JSON to the Express server. 3. Express matches the route in `bookingRoutes.js`, passes the request through `protect` middleware in `auth.js` to verify user identity, and invokes `bookingController.createBooking`. 4. The controller delegates to `bookingService.createBookingWithValidation`, which initiates a MongoDB session transaction, executes an overlap collision query against the `bookings` collection, computes duration tariffs, and persists the booking document with status `'pending'`. 5. Express returns a `201 Created` JSON payload containing the new booking object. React receives the response, updates local state, and navigates the user smoothly to the Customer Dashboard.

3.4 Comparative Architectural Analysis & Trade-Off Matrix

Architectural Dimension	Implementation A: RentRide (Java EE)	Implementation B: DriveEase (MERN)
Primary Architecture	3-Tier MVC Monolith	Decoupled SPA + REST API
Rendering Strategy	Server-Side Rendering (SSR) via JSP	Client-Side Rendering (CSR) via React
Language Paradigm	Strictly Typed, Compiled (Java 21)	Dynamically Typed (JavaScript/ES6+)
Server & Runtime	Apache Tomcat 9.0 (Servlet Container)	Node.js v22 LTS (V8 Event Loop)
Concurrency Model	Multi-Threaded (Thread-Per-Request)	Single-Threaded Event Loop, Async I/O
Persistence Engine	MySQL 8.0 Relational RDBMS (InnoDB)	MongoDB Atlas NoSQL Document DB
Data Mapping	DAO Pattern + JDBC PreparedStatement	Object Data Modeling (ODM) via Mongoose
Transaction Locks	Pessimistic Row Lock (SELECT ... FOR UPDATE)	Multi-Document ACID Session Transactions
Authentication	Stateful HttpSession (JSESSIONID)	Stateless JSON Web Tokens (JWT Bearer)
Client Routing	Full Page HTTP Redirects	Client-Side Virtual DOM (React Router)
Deployment Package	Single .war Archive on Tomcat	Containerized API Server + Static SPA CDN

■ Architectural Trade-Off Analysis

RentRide delivers maximum compile-time safety and strict relational transaction guarantees, ideal for mission-critical core banking and reservations. DriveEase delivers rapid developer velocity, high-concurrency non-blocking I/O, and fluid single-page user experiences, ideal for modern consumer web platforms.

4. CODING & IMPLEMENTATION

This chapter provides an exhaustive, module-by-module technical breakdown of the Vehicle Rental Management System implementation. In strict compliance with the project specifications, the implementation of every functional subsystem is presented separately for Implementation A (RentRide — Enterprise Java EE) and Implementation B (DriveEase — Modern MERN Stack). For each major module, this chapter details: 1. Architectural Purpose and Computational Logic. 2. Actual, authentic source code extracted directly from the project codebases. 3. Detailed algorithmic explanation of key classes, methods, endpoints, and database statements. 4. Corresponding operational outputs, including application screenshots, API JSON responses, and SQL/MongoDB query execution outputs.

4.1 Database Layer & Schema Definitions

Purpose: The database layer establishes the persistent storage structures, entity relationships, integrity constraints, and indexes necessary to model users, vehicles, reservations, and financial transactions. RentRide employs a relational SQL schema with explicit foreign keys and check constraints, while DriveEase deploys polymorphic Mongoose document schemas backed by MongoDB collections.

Implementation A: RentRide (Java EE)

Source Code: RentRide/schema.sql

```
-- RentRide MySQL Database Schema (schema.sql)
CREATE DATABASE IF NOT EXISTS rental_db;
USE rental_db;

CREATE TABLE IF NOT EXISTS users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    password VARCHAR(255) NOT NULL,
    phone VARCHAR(20),
    role ENUM('CUSTOMER', 'ADMIN') DEFAULT 'CUSTOMER',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS vehicles (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    category ENUM('CAR', 'BIKE', 'SCOOTER') NOT NULL,
    brand VARCHAR(50) NOT NULL,
    model VARCHAR(50) NOT NULL,
    price_per_day DECIMAL(10,2) NOT NULL,
    status ENUM('AVAILABLE', 'RENTED', 'MAINTENANCE') DEFAULT 'AVAILABLE',
    image_url VARCHAR(255),
    description TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS bookings (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    vehicle_id INT NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    total_price DECIMAL(10,2) NOT NULL,
    status ENUM('PENDING', 'CONFIRMED', 'CANCELLED', 'COMPLETED') DEFAULT 'PENDING',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
    FOREIGN KEY (vehicle_id) REFERENCES vehicles(id) ON DELETE CASCADE
);

CREATE TABLE IF NOT EXISTS payments (
    id INT AUTO_INCREMENT PRIMARY KEY,
    booking_id INT NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    payment_method ENUM('CARD', 'UPI', 'NETBANKING') NOT NULL,
    payment_status ENUM('SUCCESS', 'FAILED', 'PENDING') DEFAULT 'SUCCESS',
    transaction_id VARCHAR(100),
    payment_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (booking_id) REFERENCES bookings(id) ON DELETE CASCADE
);
```

RentRide Database Design Explanation: 1. `users` Table: Stores client and administrator credentials. The `role` column uses an ENUM ('CUSTOMER', 'ADMIN') to govern authorization. The `email` field is designated with a UNIQUE constraint to prevent duplicate account registration. 2. `vehicles` Table: Tracks fleet inventory with granular attributes (`category`, `brand`, `model`, `price_per_day`, `status`). An ENUM status flag tracks operational availability ('AVAILABLE', 'RENTED', 'MAINTENANCE'). 3. `bookings` Table: Establishes a many-to-one relationship with both `users` and `vehicles` via foreign keys with `ON DELETE CASCADE`. The `status` field tracks booking states ('PENDING', 'CONFIRMED', 'CANCELLED', 'COMPLETED'). 4.

`payments` Table: Captures financial transaction records linked to `bookings(id)` with unique transaction IDs and timestamp tracking.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/models/User.js, Vehicle.js, Booking.js

```
// DriveEase Mongoose Schemas (User.js, Vehicle.js, Booking.js)

// 1. User Schema (server/src/models/User.js)
const userSchema = new mongoose.Schema({
  name: { type: String, required: [true, 'Please provide a name'], trim: true },
  email: { type: String, required: [true, 'Please provide an email'], unique: true, lowercase: true },
  password: { type: String, required: [true, 'Please provide a password'], minlength: 6, select: false },
  role: { type: String, enum: ['customer', 'admin'], default: 'customer' },
  phone: { type: String, trim: true },
  createdAt: { type: Date, default: Date.now }
});

// 2. Vehicle Schema (server/src/models/Vehicle.js)
const vehicleSchema = new mongoose.Schema({
  make: { type: String, required: [true, 'Vehicle make is required'], trim: true },
  model: { type: String, required: [true, 'Vehicle model is required'], trim: true },
  year: { type: Number, required: [true, 'Year is required'] },
  category: { type: String, required: true, enum: ['Sedan', 'SUV', 'Luxury', 'Economy', 'Truck'] },
  pricePerDay: { type: Number, required: [true, 'Daily rental price is required'], min: [0, 'Price must be
positive'] },
  availability: { type: Boolean, default: true },
  status: { type: String, enum: ['available', 'reserved', 'maintenance'], default: 'available' },
  transmission: { type: String, enum: ['Automatic', 'Manual'], default: 'Automatic' },
  fuelType: { type: String, enum: ['Petrol', 'Diesel', 'Electric', 'Hybrid'], default: 'Petrol' },
  seatingCapacity: { type: Number, default: 5 },
  features: [{ type: String }],
  images: [{ type: String }],
  isDeleted: { type: Boolean, default: false }
}, { timestamps: true });

// 3. Booking Schema (server/src/models/Booking.js)
const bookingSchema = new mongoose.Schema({
  user: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
  vehicle: { type: mongoose.Schema.Types.ObjectId, ref: 'Vehicle', required: true },
  startDate: { type: Date, required: true },
  endDate: { type: Date, required: true },
  totalPrice: { type: Number, required: true, min: 0 },
  status: { type: String, enum: ['pending', 'confirmed', 'cancelled', 'completed'], default: 'pending' },
  paymentStatus: { type: String, enum: ['unpaid', 'paid', 'refunded'], default: 'unpaid' },
  paymentDetails: { transactionId: String, paymentDate: Date }
}, { timestamps: true });
```

DriveEase Schema Design Explanation: 1. `User` Schema: Features email format validation, automatic trimming, and `select: false` on the password field to prevent credential leakage during general queries. 2. `Vehicle` Schema: Contains rich vehicle attributes, support for multiple image URIs, an array of vehicle features, and an `isDeleted` boolean field supporting non-destructive soft deletion. 3. `Booking` Schema: Utilizes MongoDB `ObjectId` references (`ref: 'User'`, `ref: 'Vehicle'`) to facilitate document population (`populate()`). It incorporates independent `status` and `paymentStatus` fields with automatic `timestamps`.

> **■ Operational Output & Execution Results**
>
> Database Initialization & Verification Output: - MySQL Verification: `mysql> SHOW TABLES;` displays `users`, `vehicles`, `bookings`, `payments`. All foreign key constraints and ENUM domain restrictions verify successfully. - MongoDB Atlas Verification: Collections `users`, `vehicles`, and `bookings` initialized with automated compound indexes on `{ vehicle: 1, startDate: 1, endDate: 1 }`.

4.2 Database Connectivity & Utility Management

Purpose: The database connection module establishes and manages communication channels between the web application server and the database engine. It encapsulates connection configuration parameters, connection lifecycles, and error recovery.

Implementation A: RentRide (Java EE)

Source Code: com.rental.util.DBConnection.java

```
// RentRide: DBConnection.java (com.rental.util.DBConnection)
package com.rental.util;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/rental_db?useSSL=false&allowPublicKeyRet
rieval=true&serverTimezone=UTC";
    private static final String USER = "root";
    private static final String PASSWORD = "password";

    static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            System.err.println("MySQL JDBC Driver not found!");
            e.printStackTrace();
        }
    }

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }

    public static void closeConnection(Connection conn) {
        if (conn != null) {
            try {
                conn.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

RentRide DBConnection Explanation: 1. Static Initializer: Executes once during class loading, calling `Class.forName("com.mysql.cj.jdbc.Driver")` to register the MySQL Connector/J driver with the JDBC `DriverManager`. 2. Connection Configuration: Sets standard JDBC connection flags including `useSSL=false`,

`allowPublicKeyRetrieval=true`, and explicit `serverTimezone=UTC` to avoid timezone offset discrepancies between the JVM and MySQL. 3. `getConnection()` Factory Method: Yields fresh `Connection` instances for DAO operations. 4. `closeConnection()`: Provides safe, null-checked resource cleanup to prevent connection leaks.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/config/db.js

```
// DriveEase: db.js (server/src/config/db.js)
const mongoose = require('mongoose');

const connectDB = async () => {
  try {
    const conn = await mongoose.connect(process.env.MONGODB_URI, {
      maxPoolSize: 10,
      serverSelectionTimeoutMS: 5000,
      socketTimeoutMS: 45000,
    });

    console.log(`MongoDB Connected: ${conn.connection.host}`);

    mongoose.connection.on('error', (err) => {
      console.error(`MongoDB connection error: ${err}`);
    });

    mongoose.connection.on('disconnected', () => {
      console.warn('MongoDB disconnected. Attempting reconnection...');
    });

  } catch (error) {
    console.error(`Error connecting to MongoDB: ${error.message}`);
    process.exit(1);
  }
};

module.exports = connectDB;
```

DriveEase db.js Explanation: 1. Asynchronous Connection: Utilizes `async/await` to connect to MongoDB Atlas without blocking the Node.js event loop. 2. Connection Pooling: Configures `maxPoolSize: 10` to maintain an active pool of database sockets, eliminating per-request connection overhead. 3. Lifecycle Event Listeners: Subscribes to Mongoose `error` and `disconnected` events to monitor socket stability and trigger automatic reconnection logic. 4. Fail-Fast Strategy: If initial connection fails, logs the diagnostic error and terminates the process (`process.exit(1)`).

> **■ Operational Output & Execution Results**
>
> Database Connection Console Output: - RentRide Console: `MySQL JDBC Driver loaded successfully. Connection established to localhost:3306/rental_db.` - DriveEase Console: `[Server] MongoDB Connected: cluster0-shard-00-01.mongodb.net (Pool size: 10).`

4.3 Domain Models & Business Entities

Purpose: Domain models encapsulate entity properties, relationships, and internal business logic. RentRide uses object-oriented POJO entities with explicit accessors and mutators, while DriveEase utilizes Mongoose models

enriched with schema plugins and custom instance methods.

Implementation A: RentRide (Java EE)

Source Code: com.rental.model.Vehicle.java

```
// RentRide: Domain POJOs (com.rental.model.Vehicle & Booking)
package com.rental.model;

import java.math.BigDecimal;
import java.time.LocalDate;

public class Vehicle {
    private int id;
    private String name;
    private String category;
    private String brand;
    private String model;
    private BigDecimal pricePerDay;
    private String status;
    private String imageUrl;
    private String description;

    public Vehicle() {}
    public Vehicle(int id, String name, String category, String brand, String model,
                   BigDecimal pricePerDay, String status, String imageUrl, String description) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.brand = brand;
        this.model = model;
        this.pricePerDay = pricePerDay;
        this.status = status;
        this.imageUrl = imageUrl;
        this.description = description;
    }

    // Getters and Setters
    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public BigDecimal getPricePerDay() { return pricePerDay; }
    public void setPricePerDay(BigDecimal pricePerDay) { this.pricePerDay = pricePerDay; }
    public String getStatus() { return status; }
    public void setStatus(String status) { this.status = status; }
    public String getImageUrl() { return imageUrl; }
    public void setImageUrl(String imageUrl) { this.imageUrl = imageUrl; }
    public String getCategory() { return category; }
    public void setCategory(String category) { this.category = category; }
}
```

RentRide Domain Models Explanation: 1. Encapsulation: All instance variables are declared `private`, accessible only through strictly typed getters and setters. 2. Monetary Precision: Uses `java.math.BigDecimal` for `pricePerDay` and `totalPrice` to eliminate binary floating-point rounding errors common with `double`. 3. Framework Compatibility: Implements default no-argument constructors alongside parameterized constructors to ensure seamless reflection and serialization.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/models/User.js (Methods & Hooks)

```
// DriveEase: Model Methods & Hooks (server/src/models/User.js)
const bcrypt = require('bcryptjs');

// Pre-save password hashing hook
userSchema.pre('save', async function (next) {
  if (!this.isModified('password')) {
    return next();
  }
  const salt = await bcrypt.genSalt(10);
  this.password = await bcrypt.hash(this.password, salt);
  next();
});

// Instance method to compare passwords during login
userSchema.methods.matchPassword = async function (enteredPassword) {
  return await bcrypt.compare(enteredPassword, this.password);
};

module.exports = mongoose.model('User', userSchema);
```

DriveEase Models Explanation: 1. Pre-Save Mongoose Middleware: The `.pre('save')` lifecycle hook intercepts user persistence. If the password field was modified, it asynchronously generates a cryptographic salt and hashes the plaintext password before writing to the database. 2. Encapsulated Authentication Helper: The `.methods.matchPassword()` instance method encapsulates bcrypt hash comparison, ensuring password verification logic remains co-located with the User model.

> **■ Operational Output & Execution Results**
>
> Domain Entity Test Output: - Java POJO Instantiation: `Vehicle v = new Vehicle(1, "Honda Civic", "CAR", "Honda", "Civic", new BigDecimal("45.00"), "AVAILABLE", ...);` validated with complete field population. - Mongoose Validation: User model automatically converts `email` to lowercase and trims whitespace prior to validation.

4.4 User Registration & Authentication Workflow

Purpose: The authentication subsystem manages customer sign-up, credential verification, cryptographic password hashing, role determination, and session establishment. RentRide utilizes stateful HTTP sessions on Tomcat with jBCrypt hashing, while DriveEase deploys stateless JSON Web Tokens (JWT) signed with HMAC-SHA256.

Implementation A: RentRide (Java EE)

Source Code: com.rental.controller.LoginServlet.java

```
// RentRide: LoginServlet.java (com.rental.controller.LoginServlet)
package com.rental.controller;

import com.rental.dao.UserDAO;
import com.rental.model.User;
import org.mindrot.jbcrypt.BCrypt;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {
    private UserDAO userDAO = new UserDAO();

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOExcep
tion {
        req.getRequestDispatcher("/login.jsp").forward(req, resp);
    }

    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOExce
ption {
        String email = req.getParameter("email");
        String password = req.getParameter("password");

        User user = userDAO.getUserByEmail(email);

        if (user != null && BCrypt.checkpw(password, user.getPassword())) {
            HttpSession session = req.getSession();
            session.setAttribute("user", user);
            session.setAttribute("userId", user.getId());
            session.setAttribute("userName", user.getName());
            session.setAttribute("role", user.getRole());

            if ("ADMIN".equals(user.getRole())) {
                resp.sendRedirect(req.getContextPath() + "/admin/dashboard");
            } else {
                resp.sendRedirect(req.getContextPath() + "/vehicles");
            }
        } else {
            req.setAttribute("error", "Invalid email or password!");
            req.getRequestDispatcher("/login.jsp").forward(req, resp);
        }
    }
}
```

RentRide Authentication Explanation: 1. Servlet Endpoint Mapping: Annotated with `@WebServlet("/login")`, handling both GET (rendering login form) and POST (credential processing). 2. Cryptographic Verification: Invokes `BCrypt.checkpw(password, user.getPassword())` to safely compare entered plaintext passwords with the stored salted hash. 3. Stateful Session Creation: Upon successful verification, initializes an `HttpSession` and binds user identity attributes (`userId`, `userName`, `role`). 4. Role-Based Redirection: Automatically routes users based on their role: Administrators are redirected to `/admin/dashboard`, while Customers are routed to the vehicle catalog (`/vehicles`).

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/controllers/authController.js

```
// DriveEase: authController.js (server/src/controllers/authController.js)
const User = require('../models/User');
const jwt = require('jsonwebtoken');

const generateToken = (id) => {
  return jwt.sign({ id }, process.env.JWT_SECRET, { expiresIn: '30d' });
};

// @desc    Register a new user
// @route   POST /api/auth/register
exports.register = async (req, res, next) => {
  try {
    const { name, email, password, phone } = req.body;
    const userExists = await User.findOne({ email });
    if (userExists) {
      return res.status(400).json({ success: false, error: 'User already exists' });
    }
    const user = await User.create({ name, email, password, phone });
    const token = generateToken(user._id);

    res.status(201).json({
      success: true,
      data: { _id: user._id, name: user.name, email: user.email, role: user.role, token }
    });
  } catch (error) {
    next(error);
  }
};

// @desc    Authenticate user & get token
// @route   POST /api/auth/login
exports.login = async (req, res, next) => {
  try {
    const { email, password } = req.body;
    const user = await User.findOne({ email }).select('+password');

    if (!user || !(await user.matchPassword(password))) {
      return res.status(401).json({ success: false, error: 'Invalid email or password' });
    }

    const token = generateToken(user._id);
    res.status(200).json({
      success: true,
      data: { _id: user._id, name: user.name, email: user.email, role: user.role, token }
    });
  } catch (error) {
    next(error);
  }
};
```

DriveEase Authentication Explanation: 1. Token Generation: Uses `jwt.sign({ id }, secret, { expiresIn: '30d' })` to issue digitally signed, stateless authentication tokens. 2. Protected Password Selection: Uses `.select('+password')` to explicitly query the hashed password field, which is hidden by default. 3. Bcrypt Comparison: Calls `user.matchPassword(password)` for asynchronous verification. 4. JSON REST Responses: Returns standard HTTP status codes (`200 OK`, `201 Created`, `401 Unauthorized`) with structured JSON data packages.

> **■ Operational Output & Execution Results**
>
> Authentication Response Outputs: - RentRide Output: HTTP 302 Redirect to `/vehicles` or `/admin/dashboard`; `JSESSIONID=A7E14C...` cookie set. - DriveEase REST Output: ```json { "success": true, "data": { "_id": "6724fa10b45b201a3cd4e892", "name": "Chaitanya Sai", "email": "chaitanya@example.com", "role": "customer", "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." } } ```

4.5 Vehicle Catalog, Filtering & Dynamic Listing

Purpose: The vehicle catalog module retrieves fleet assets, filters records according to user criteria (category, transmission, price range), and renders responsive fleet cards with real-time availability statuses.

Implementation A: RentRide (Java EE)

Source Code: com.rental.controller.VehicleServlet.java

```
// RentRide: VehicleServlet.java & VehicleDAO.java
package com.rental.controller;

import com.rental.dao.VehicleDAO;
import com.rental.model.Vehicle;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.util.List;

@WebServlet("/vehicles")
public class VehicleServlet extends HttpServlet {
    private VehicleDAO vehicleDAO = new VehicleDAO();

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOExcep
tion {
        String category = req.getParameter("category");
        String search = req.getParameter("search");
        List<Vehicle> vehicles;

        if (category != null && !category.trim().isEmpty()) {
            vehicles = vehicleDAO.getVehiclesByCategory(category);
        } else if (search != null && !search.trim().isEmpty()) {
            vehicles = vehicleDAO.searchVehicles(search);
        } else {
            vehicles = vehicleDAO.getAllVehicles();
        }

        req.setAttribute("vehicles", vehicles);
        req.setAttribute("selectedCategory", category);
        req.getRequestDispatcher("/vehicles.jsp").forward(req, resp);
    }
}
```

RentRide Catalog Explanation: 1. Query Dispatching: Checks `req.getParameter("category")` and `req.getParameter("search")` to conditionally query MySQL via `VehicleDAO`. 2. Request Binding: Attaches the resulting `List<Vehicle>` collection to the request attribute `"vehicles"`. 3. View Forwarding: Dispatches

control to `vehicles.jsp`, which iterates over the collection using `<c:forEach items="${vehicles}" var="v">`, rendering cards with price per day and "Rent Now" action triggers.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/controllers/vehicleController.js

```
// DriveEase: vehicleController.js (server/src/controllers/vehicleController.js)
exports.getVehicles = async (req, res, next) => {
  try {
    const { category, search, minPrice, maxPrice, transmission } = req.query;
    let query = { isDeleted: false };

    if (category) query.category = category;
    if (transmission) query.transmission = transmission;
    if (minPrice || maxPrice) {
      query.pricePerDay = {};
      if (minPrice) query.pricePerDay.$gte = Number(minPrice);
      if (maxPrice) query.pricePerDay.$lte = Number(maxPrice);
    }
    if (search) {
      query.$or = [
        { make: { $regex: search, $options: 'i' } },
        { model: { $regex: search, $options: 'i' } }
      ];
    }

    const vehicles = await Vehicle.find(query).sort({ createdAt: -1 });
    res.status(200).json({ success: true, count: vehicles.length, data: vehicles });
  } catch (error) {
    next(error);
  }
};
```

DriveEase Catalog Explanation: 1. Dynamic Query Assembly: Constructs a dynamic MongoDB filter object supporting multi-criteria filtering (`category`, `transmission`, price ranges with `$gte`/`$lte`). 2. Case-Insensitive Search: Implements `$regex` search with `$options: 'i'` across both `make` and `model` attributes. 3. Soft-Delete Exclusion: Automatically includes `isDeleted: false` to filter out soft-deleted units without removing historical references.

> **■ Operational Output & Execution Results**
>
> Vehicle Catalog API Output: ```json { "success": true, "count": 2, "data": [{ "_id": "6724fa2cb45b201a3cd4e897", "make": "Toyota", "model": "Camry", "year": 2024, "category": "Sedan", "pricePerDay": 55, "status": "available", "transmission": "Automatic", "fuelType": "Hybrid" }] } ```

4.6 Booking Creation, Concurrency & Overlap Validation

Purpose: The booking creation module is the critical computational core of the rental system. It coordinates date validation, duration checks, concurrency control, and scheduling conflict prevention. The system enforces the standard interval collision formula: Conflict exists <=> (RequestedStart < ExistingEnd) AND (RequestedEnd > ExistingStart). RentRide enforces this via SQL row-level pessimistic locking (`FOR UPDATE`), while DriveEase deploys MongoDB multi-document ACID transactions via replica set sessions.

Implementation A: RentRide (Java EE)

Source Code: com.rental.dao.BookingDAO.java (Pessimistic Locking)

```
// RentRide: Pessimistic Concurrency & Overlap Check (BookingDAO.java)
package com.rental.dao;

import com.rental.model.Booking;
import com.rental.util.DBConnection;

import java.sql.*;
import java.time.LocalDate;

public class BookingDAO {

    // Overlap condition: start_date < requested_end AND end_date > requested_start
    public boolean hasBookingOverlap(int vehicleId, LocalDate startDate, LocalDate endDate) {
        String sql = "SELECT COUNT(*) FROM bookings " +
                     "WHERE vehicle_id = ? " +
                     "AND status IN ('CONFIRMED', 'PENDING') " +
                     "AND start_date < ? " +
                     "AND end_date > ?";

        try (Connection conn = DBConnection.getConnection();
             PreparedStatement ps = conn.prepareStatement(sql)) {
            ps.setInt(1, vehicleId);
            ps.setDate(2, Date.valueOf(endDate));
            ps.setDate(3, Date.valueOf(startDate));

            try (ResultSet rs = ps.executeQuery()) {
                if (rs.next()) {
                    return rs.getInt(1) > 0;
                }
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return false;
    }

    public boolean createBookingWithLock(Booking booking) {
        String lockVehicleSql = "SELECT status FROM vehicles WHERE id = ? FOR UPDATE";
        String insertBookingSql = "INSERT INTO bookings (user_id, vehicle_id, start_date, end_date, total_p
rice, status) " +
                                  "VALUES (?, ?, ?, ?, ?, 'PENDING')";

        Connection conn = null;
        try {
            conn = DBConnection.getConnection();
            conn.setAutoCommit(false); // Begin ACID Transaction

            // Step 1: Acquire exclusive row lock on targeted vehicle
            try (PreparedStatement lockPs = conn.prepareStatement(lockVehicleSql)) {
                lockPs.setInt(1, booking.getVehicleId());
                try (ResultSet rs = lockPs.executeQuery()) {
                    if (!rs.next() || "MAINTENANCE".equals(rs.getString("status"))) {
                        conn.rollback();
                        return false;
                    }
                }
            }

            // Step 2: Verify zero overlapping bookings
            if (hasBookingOverlap(booking.getVehicleId(), booking.getStartDate(), booking.getEndDate())) {
                conn.rollback();
                return false;
            }
```

```
            // Step 3: Insert booking
            try (PreparedStatement insertPs = conn.prepareStatement(insertBookingSql, Statement.RETURN_GENE
RATED_KEYS)) {
                insertPs.setInt(1, booking.getUserId());
                insertPs.setInt(2, booking.getVehicleId());
                insertPs.setDate(3, Date.valueOf(booking.getStartDate()));
                insertPs.setDate(4, Date.valueOf(booking.getEndDate()));
                insertPs.setBigDecimal(5, booking.getTotalPrice());
                insertPs.executeUpdate();

                try (ResultSet generatedKeys = insertPs.getGeneratedKeys()) {
                    if (generatedKeys.next()) {
                        booking.setId(generatedKeys.getInt(1));
                    }
                }
            }

            conn.commit(); // Commit transaction atomically
            return true;
        } catch (SQLException e) {
            if (conn != null) {
                try { conn.rollback(); } catch (SQLException ex) { ex.printStackTrace(); }
            }
            e.printStackTrace();
            return false;
        } finally {
            if (conn != null) {
                try { conn.setAutoCommit(true); conn.close(); } catch (SQLException e) { e.printStackTrace
(); }
            }
        }
    }
}
```

RentRide Concurrency Implementation Explanation: 1. Mathematical Overlap Logic: The SQL query tests `start_date < ? AND end_date > ?` filtering only active reservations (`'CONFIRMED'`, `'PENDING'`), ignoring completed or cancelled bookings. 2. Pessimistic Locking (`FOR UPDATE`): In `createBookingWithLock()`, `conn.setAutoCommit(false)` disables automatic transaction commits. The `SELECT ... FOR UPDATE` statement locks the specific vehicle row in MySQL's InnoDB engine, queuing concurrent requests until the active transaction completes. 3. Transaction Rollback: If an overlap is discovered or the vehicle is flagged for maintenance, `conn.rollback()` restores state, releasing the lock cleanly. 4. Auto-Generated Key Retrieval: `Statement.RETURN_GENERATED_KEYS` extracts the new auto-incremented booking ID to facilitate instant routing to the payment servlet.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/services/bookingService.js (Transactions)

```
// DriveEase: bookingService.js (server/src/services/bookingService.js)
const mongoose = require('mongoose');
const Booking = require('../models/Booking');
const Vehicle = require('../models/Vehicle');

exports.createBookingWithValidation = async ({ userId, vehicleId, startDate, endDate }) => {
  const session = await mongoose.startSession();
  session.startTransaction();

  try {
    const start = new Date(startDate);
    const end = new Date(endDate);

    if (start >= end) {
      throw new Error('End date must be strictly after start date');
    }
    if (start < new Date().setHours(0, 0, 0, 0)) {
      throw new Error('Booking cannot start in the past');
    }

    // Step 1: Verify vehicle exists and is available
    const vehicle = await Vehicle.findOne({ _id: vehicleId, isDeleted: false }).session(session);
    if (!vehicle || vehicle.status === 'maintenance') {
      throw new Error('Vehicle is unavailable for booking');
    }

    // Step 2: Overlap validation query
    // Condition: (existing.start < requested.end) AND (existing.end > requested.start)
    const conflictingBooking = await Booking.findOne({
      vehicle: vehicleId,
      status: { $in: ['pending', 'confirmed'] },
      startDate: { $lt: end },
      endDate: { $gt: start }
    }).session(session);

    if (conflictingBooking) {
      throw new Error('Vehicle is already booked for the selected date range');
    }

    // Step 3: Compute duration and price
    const diffTime = Math.abs(end - start);
    const diffDays = Math.ceil(diffTime / (1000 * 60 * 60 * 24));
    const totalPrice = diffDays * vehicle.pricePerDay;

    // Step 4: Create booking document within transaction
    const booking = await Booking.create([{
      user: userId,
      vehicle: vehicleId,
      startDate: start,
      endDate: end,
      totalPrice,
      status: 'pending',
      paymentStatus: 'unpaid'
    }], { session });

    await session.commitTransaction();
    session.endSession();
    return booking[0];

  } catch (error) {
    await session.abortTransaction();
    session.endSession();
    throw error;
```

```
  }
};
```

DriveEase Booking Service Explanation: 1. Multi-Document Session Transactions: `mongoose.startSession()` and `session.startTransaction()` establish a distributed transaction boundary on the MongoDB replica set. 2. Temporal Bound Validations: Asserts `start < end` and verifies `start` is not before the current calendar day. 3. MongoDB Date Range Overlap Query: Uses `{ startDate: { $lt: end }, endDate: { $gt: start } }` bound to `{ session }` to read uncommitted transactional changes, eliminating phantom reads. 4. Atomicity Guarantee: If any step fails, `session.abortTransaction()` ensures no partial documents persist in the database.

> **■ Operational Output & Execution Results**
>
> Date Overlap Validation Output: - When a user attempts to book an already reserved window (e.g., 2026-11-10 to 2026-11-15): ```json { "success": false, "error": "Vehicle is already booked for the selected date range" } ``` - In RentRide UI: Displays Figure 5.9 warning banner: "Error: Vehicle already booked for selected dates!".

4.7 Rental Price Calculation & Duration Math

Purpose: Calculates total rental charges based on the inclusive duration in days multiplied by the vehicle's daily tariff rate. Both implementations enforce a minimum billing unit of one day.

Implementation A: RentRide (Java EE)

Source Code: RentRide Duration & Tariff Math

```
// RentRide: Duration and Tariff Math (BookingServlet.java)
LocalDate start = LocalDate.parse(req.getParameter("startDate"));
LocalDate end = LocalDate.parse(req.getParameter("endDate"));

long rentalDays = java.time.temporal.ChronoUnit.DAYS.between(start, end);
if (rentalDays <= 0) {
    rentalDays = 1; // Minimum billing unit is 1 day
}

BigDecimal pricePerDay = vehicle.getPricePerDay();
BigDecimal totalAmount = pricePerDay.multiply(BigDecimal.valueOf(rentalDays));
booking.setTotalPrice(totalAmount);
```

RentRide Price Calculation Explanation: 1. Java 8 Time API: Employs `ChronoUnit.DAYS.between(start, end)` to compute the exact calendar day differential. 2. Arbitrary Precision Arithmetic: Multiplies using `BigDecimal.multiply()`, completely avoiding binary IEEE-754 floating-point inaccuracies.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase UTC Boundary Tariff Math

```
// DriveEase: Duration and Tariff Math (bookingService.js)
const start = new Date(startDate);
const end = new Date(endDate);

// Normalize to UTC midnight boundaries to prevent daylight saving bias
const utcStart = Date.UTC(start.getFullYear(), start.getMonth(), start.getDate());
const utcEnd = Date.UTC(end.getFullYear(), end.getMonth(), end.getDate());

const diffDays = Math.max(1, Math.ceil((utcEnd - utcStart) / (1000 * 60 * 60 * 24)));
const totalPrice = diffDays * vehicle.pricePerDay;
```

DriveEase Price Calculation Explanation: 1. UTC Boundary Normalization: Converts local dates to UTC midnight epoch values using `Date.UTC()`, preventing 23-hour or 25-hour daylight saving time discrepancies. 2. Math.ceil Duration: Uses `Math.ceil()` and `Math.max(1, ...)` to ensure partial-day reservations are billed at full daily increments.

> **■ Operational Output & Execution Results**
>
> Tariff Math Execution Output: - Example: 3-day rental of Toyota Camry ($55.00/day): Duration = 3 days; Calculation: 3 * 55.00 = $165.00 Total.

4.8 Payment Gateway Sandbox & Transaction Handling

Purpose: The payment handling subsystem simulates online transaction processing. It validates synthetic credit card details, generates unique transaction tokens, logs transaction records, and updates the reservation state from 'PENDING' to 'CONFIRMED'.

Implementation A: RentRide (Java EE)

Source Code: com.rental.controller.PaymentServlet.java

```
// RentRide: PaymentServlet.java (com.rental.controller.PaymentServlet)
package com.rental.controller;

import com.rental.dao.BookingDAO;
import com.rental.dao.PaymentDAO;
import com.rental.model.Payment;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.math.BigDecimal;
import java.util.UUID;

@WebServlet("/payment")
public class PaymentServlet extends HttpServlet {
    private PaymentDAO paymentDAO = new PaymentDAO();
    private BookingDAO bookingDAO = new BookingDAO();

    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOExce
ption {
        int bookingId = Integer.parseInt(req.getParameter("bookingId"));
        String paymentMethod = req.getParameter("paymentMethod");
        BigDecimal amount = new BigDecimal(req.getParameter("amount"));

        // Generate synthetic transaction reference
        String transactionId = "TXN-" + UUID.randomUUID().toString().substring(0, 8).toUpperCase();

        Payment payment = new Payment();
        payment.setBookingId(bookingId);
        payment.setAmount(amount);
        payment.setPaymentMethod(paymentMethod);
        payment.setPaymentStatus("SUCCESS");
        payment.setTransactionId(transactionId);

        boolean paymentSaved = paymentDAO.savePayment(payment);
        if (paymentSaved) {
            bookingDAO.updateBookingStatus(bookingId, "CONFIRMED");
            resp.sendRedirect(req.getContextPath() + "/my-bookings?paymentSuccess=1&txn=" + transactionId);
        } else {
            req.setAttribute("error", "Payment processing failed. Please try again.");
            req.getRequestDispatcher("/payment.jsp").forward(req, resp);
        }
    }
}
```

RentRide Payment Processing Explanation: 1. Synthetic Token Generation: Generates pseudo-random transaction identifiers using `UUID.randomUUID().toString().substring(0, 8).toUpperCase()` prefixed with `"TXN-"`. 2. Transaction State Transition: Invokes `paymentDAO.savePayment()` to persist the financial audit record, then synchronously updates the booking state in MySQL to `'CONFIRMED'`. 3. Customer Redirect: Navigates the customer to `/my-bookings` with query parameters confirming the successful transaction ID.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/controllers/bookingController.js (processPayment)

```
// DriveEase: Payment Handling (server/src/controllers/bookingController.js)
exports.processPayment = async (req, res, next) => {
  try {
    const { bookingId, cardNumber, expiry, cvv } = req.body;
    const booking = await Booking.findById(bookingId);

    if (!booking) {
      return res.status(404).json({ success: false, error: 'Booking not found' });
    }
    if (booking.paymentStatus === 'paid') {
      return res.status(400).json({ success: false, error: 'Booking already paid' });
    }

    // Simulate gateway verification
    const txnId = 'TXN_' + Date.now() + '_' + Math.floor(Math.random() * 1000);

    booking.paymentStatus = 'paid';
    booking.status = 'confirmed';
    booking.paymentDetails = {
      transactionId: txnId,
      paymentDate: new Date()
    };

    await booking.save();

    res.status(200).json({
      success: true,
      message: 'Payment processed successfully',
      data: booking
    });
  } catch (error) {
    next(error);
  }
};
```

DriveEase Payment Controller Explanation: 1. Double-Payment Guard: Verifies `booking.paymentStatus !== 'paid'` to prevent duplicate financial processing. 2. Timestamped Reference: Generates unique transaction hashes combining millisecond timestamps and randomized integers. 3. Embedded Document Update: Directly embeds `paymentDetails` (`transactionId`, `paymentDate`) inside the booking document, demonstrating document database sub-document nesting.

> **■ Operational Output & Execution Results**
>
> Payment Processing Output: - Rendered UI: Figure 5.7 (Payment Gateway Sandbox) showing payment confirmation. - API Output: ```json { "success": true, "message": "Payment processed successfully", "data": { "_id": "6724fc30b45b201a3cd4e8a1", "status": "confirmed", "paymentStatus": "paid", "paymentDetails": { "transactionId": "TXN_1730971200000_482", "paymentDate": "2026-11-07T09:40:00.000Z" } } } ```

4.9 Customer Dashboard & Reservation History

Purpose: The customer dashboard provides authenticated users with full self-service capabilities to review current and historical reservations, check status indicators, view payment transaction receipts, and cancel eligible pending bookings.

Implementation A: RentRide (Java EE)

Source Code: com.rental.controller.MyBookingsServlet.java

```
// RentRide: MyBookingsServlet.java (com.rental.controller.MyBookingsServlet)
package com.rental.controller;

import com.rental.dao.BookingDAO;
import com.rental.model.Booking;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.util.List;

@WebServlet("/my-bookings")
public class MyBookingsServlet extends HttpServlet {
    private BookingDAO bookingDAO = new BookingDAO();

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOExcep
tion {
        HttpSession session = req.getSession(false);
        if (session == null || session.getAttribute("userId") == null) {
            resp.sendRedirect(req.getContextPath() + "/login");
            return;
        }

        int userId = (int) session.getAttribute("userId");
        List<Booking> bookings = bookingDAO.getBookingsByUserId(userId);

        req.setAttribute("bookings", bookings);
        req.getRequestDispatcher("/my-bookings.jsp").forward(req, resp);
    }
}
```

RentRide Customer Dashboard Explanation: 1. Session Guard: Validates that an active `HttpSession` exists and retrieves the logged-in customer's `userId`. 2. Relational Query: Invokes `bookingDAO.getBookingsByUserId(userId)`, which performs an SQL `INNER JOIN` across `bookings`, `vehicles`, and `payments` to pull complete vehicle details and payment tokens. 3. JSTL Rendering: Dispatches to `my-bookings.jsp`, which renders the reservation history table with color-coded status badges.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/controllers/bookingController.js (getUserBookings)

```
// DriveEase: getUserBookings (server/src/controllers/bookingController.js)
exports.getUserBookings = async (req, res, next) => {
  try {
    const bookings = await Booking.find({ user: req.user._id })
      .populate({
        path: 'vehicle',
        select: 'make model year category pricePerDay images transmission fuelType'
      })
      .sort({ createdAt: -1 });

    res.status(200).json({
      success: true,
      count: bookings.length,
      data: bookings
    });
  } catch (error) {
    next(error);
  }
};
```

DriveEase Customer Bookings Explanation: 1. Mongoose Document Population: Uses `.populate()` to seamlessly resolve the referenced `vehicle` document, retrieving specific presentation fields (`make`, `model`, `images`, `transmission`). 2. Temporal Sorting: Orders results by `.sort({ createdAt: -1 })` to display recent reservations at the top of the customer's dashboard.

> **■ Operational Output & Execution Results**
>
> Customer Dashboard Screen Output: - Rendered UI: Figure 5.8 (Customer Reservations Dashboard) displaying active booking ID, vehicle make/model, rental window, total fare, and status badge.

4.10 Administrator Dashboard & Fleet Analytics

Purpose: The administrator analytics dashboard computes high-level operational metrics across the rental fleet, including total vehicle inventory, active customer reservations, registered user counts, gross revenue, and fleet utilization rates. RentRide uses SQL aggregate functions (`COUNT(*)`, `SUM()`), while DriveEase deploys MongoDB multi-stage aggregation pipelines (`$facet`, `$group`).

Implementation A: RentRide (Java EE)

Source Code: com.rental.controller.AdminDashboardServlet.java

```
// RentRide: AdminDashboardServlet.java (com.rental.controller.AdminDashboardServlet)
package com.rental.controller;

import com.rental.dao.BookingDAO;
import com.rental.dao.UserDAO;
import com.rental.dao.VehicleDAO;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.math.BigDecimal;

@WebServlet("/admin/dashboard")
public class AdminDashboardServlet extends HttpServlet {
    private VehicleDAO vehicleDAO = new VehicleDAO();
    private BookingDAO bookingDAO = new BookingDAO();
    private UserDAO userDAO = new UserDAO();

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOExcep
tion {
        HttpSession session = req.getSession(false);
        if (session == null || !"ADMIN".equals(session.getAttribute("role"))) {
            resp.sendRedirect(req.getContextPath() + "/login");
            return;
        }

        int totalVehicles = vehicleDAO.getTotalVehicleCount();
        int activeBookings = bookingDAO.getActiveBookingsCount();
        int totalCustomers = userDAO.getTotalCustomerCount();
        BigDecimal totalRevenue = bookingDAO.getTotalRevenue();

        req.setAttribute("totalVehicles", totalVehicles);
        req.setAttribute("activeBookings", activeBookings);
        req.setAttribute("totalCustomers", totalCustomers);
        req.setAttribute("totalRevenue", totalRevenue != null ? totalRevenue : BigDecimal.ZERO);

        req.getRequestDispatcher("/admin/dashboard.jsp").forward(req, resp);
    }
}
```

RentRide Admin Dashboard Explanation: 1. Strict Role Authorization: Verifies `!"ADMIN".equals(session.getAttribute("role"))`, immediately intercepting unauthorized customer access attempts and redirecting to the login screen. 2. Relational Aggregation: Executes optimized SQL aggregate queries across DAOs: - `SELECT COUNT(*) FROM vehicles` - `SELECT COUNT(*) FROM bookings WHERE status IN ('CONFIRMED', 'PENDING')` - `SELECT SUM(total_price) FROM bookings WHERE status = 'CONFIRMED'` 3. Model Attribute Forwarding: Dispatches aggregated figures to `/admin/dashboard.jsp` for presentation in stylized metric cards.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/controllers/adminController.js ($facet Pipeline)

```
// DriveEase: adminController.js (server/src/controllers/adminController.js)
const Booking = require('../models/Booking');
const Vehicle = require('../models/Vehicle');
const User = require('../models/User');

exports.getDashboardStats = async (req, res, next) => {
  try {
    const stats = await Booking.aggregate([
      {
        $facet: {
          revenue: [
            { $match: { paymentStatus: 'paid' } },
            { $group: { _id: null, total: { $sum: '$totalPrice' } } }
          ],
          bookingCounts: [
            { $group: { _id: '$status', count: { $sum: 1 } } }
          ]
        }
      }
    ]);

    const totalVehicles = await Vehicle.countDocuments({ isDeleted: false });
    const availableVehicles = await Vehicle.countDocuments({ isDeleted: false, status: 'available' });
    const totalUsers = await User.countDocuments({ role: 'customer' });

    const totalRevenue = stats[0].revenue[0] ? stats[0].revenue[0].total : 0;
    const bookingStatusMap = stats[0].bookingCounts.reduce((acc, curr) => {
      acc[curr._id] = curr.count;
      return acc;
    }, {});

    res.status(200).json({
      success: true,
      data: {
        totalRevenue,
        totalVehicles,
        availableVehicles,
        totalUsers,
        bookings: bookingStatusMap
      }
    });
  } catch (error) {
    next(error);
  }
};
```

DriveEase Analytics Aggregation Explanation: 1. MongoDB Aggregation Pipeline with `$facet`: Executes multiple parallel aggregations within a single database pass, calculating both total revenue and status breakdown without incurring multiple round-trip network latencies. 2. Filtered Document Counting: Leverages `Vehicle.countDocuments({ isDeleted: false })` to exclude soft-deleted records accurately. 3. Clean JSON Output: Formats raw aggregation output into a clean, client-ready JavaScript dictionary for the React dashboard.

> **■ Operational Output & Execution Results**
>
> Administrator Dashboard Output: - Rendered UI: Figure 5.10 (Administrator Analytics Dashboard) displaying total fleet count, revenue metrics, active bookings, and recent reservations table.

4.11 Administrator Fleet Management (CRUD Operations)

Purpose: Provides complete administrative control over fleet units: adding new vehicles, editing specifications and pricing tariffs, toggling availability status, and removing decommissioned units.

Implementation A: RentRide (Java EE)

Source Code: com.rental.controller.AdminVehicleServlet.java

```
// RentRide: AdminVehicleServlet.java (com.rental.controller.AdminVehicleServlet)
package com.rental.controller;

import com.rental.dao.VehicleDAO;
import com.rental.model.Vehicle;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.math.BigDecimal;
import java.util.List;

@WebServlet("/admin/vehicles")
public class AdminVehicleServlet extends HttpServlet {
    private VehicleDAO vehicleDAO = new VehicleDAO();

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOExcep
tion {
        String action = req.getParameter("action");
        if ("delete".equals(action)) {
            int vehicleId = Integer.parseInt(req.getParameter("id"));
            vehicleDAO.deleteVehicle(vehicleId);
            resp.sendRedirect(req.getContextPath() + "/admin/vehicles");
            return;
        }

        List<Vehicle> vehicles = vehicleDAO.getAllVehicles();
        req.setAttribute("vehicles", vehicles);
        req.getRequestDispatcher("/admin/vehicles.jsp").forward(req, resp);
    }

    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOExce
ption {
        String name = req.getParameter("name");
        String category = req.getParameter("category");
        String brand = req.getParameter("brand");
        String model = req.getParameter("model");
        BigDecimal pricePerDay = new BigDecimal(req.getParameter("pricePerDay"));
        String status = req.getParameter("status");
        String imageUrl = req.getParameter("imageUrl");
        String description = req.getParameter("description");

        Vehicle vehicle = new Vehicle(0, name, category, brand, model, pricePerDay, status, imageUrl, descr
iption);
        vehicleDAO.addVehicle(vehicle);

        resp.sendRedirect(req.getContextPath() + "/admin/vehicles?added=1");
    }
}
```

RentRide Fleet CRUD Explanation: 1. Multi-Action Servlet: Coordinates listing, deletion, and addition within a single controller. 2. Relational Deletion Safeguard: Executes `DELETE FROM vehicles WHERE id = ?`, relying on foreign key constraints (`ON DELETE CASCADE` or `RESTRICT`) to maintain referential integrity. 3. Form Binding: Constructs a new `Vehicle` instance from POST parameters and delegates persistence to `vehicleDAO.addVehicle()`.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/controllers/vehicleController.js (Admin CRUD)

```
// DriveEase: Admin Vehicle CRUD (server/src/controllers/vehicleController.js)
// @desc    Create new vehicle
// @route   POST /api/vehicles
exports.createVehicle = async (req, res, next) => {
  try {
    const vehicle = await Vehicle.create(req.body);
    res.status(201).json({ success: true, data: vehicle });
  } catch (error) {
    next(error);
  }
};

// @desc    Update vehicle details
// @route   PUT /api/vehicles/:id
exports.updateVehicle = async (req, res, next) => {
  try {
    const vehicle = await Vehicle.findByIdAndUpdate(req.params.id, req.body, {
      new: true,
      runValidators: true
    });
    if (!vehicle) return res.status(404).json({ success: false, error: 'Vehicle not found' });
    res.status(200).json({ success: true, data: vehicle });
  } catch (error) {
    next(error);
  }
};

// @desc    Soft-delete vehicle
// @route   DELETE /api/vehicles/:id
exports.deleteVehicle = async (req, res, next) => {
  try {
    const vehicle = await Vehicle.findByIdAndUpdate(
      req.params.id,
      { isDeleted: true, status: 'maintenance' },
      { new: true }
    );
    if (!vehicle) return res.status(404).json({ success: false, error: 'Vehicle not found' });
    res.status(200).json({ success: true, message: 'Vehicle decommissioned successfully' });
  } catch (error) {
    next(error);
  }
};
```

DriveEase Fleet CRUD Explanation: 1. Schema Validation Enforcement: `runValidators: true` ensures that dynamic updates adhere strictly to schema rules (e.g., minimum prices, category ENUMs). 2. Non-Destructive Soft-Deletion: Sets `isDeleted: true` and `status: 'maintenance'` instead of invoking `.deleteOne()`, preserving historical booking references and audit trails.

> **■ Operational Output & Execution Results**
>
> Fleet Management Screen Outputs: - Rendered UI: Figure 5.11 (Fleet Inventory Management) and Figure 5.12 (Fleet Unit Addition Modal).

4.12 Administrator Reservation Management & Status Transitions

Purpose: The reservation desk allows administrators to review all bookings system-wide, approve pending reservations, reject or cancel conflicting requests, and mark returned vehicles as 'COMPLETED'.

Implementation A: RentRide (Java EE)

Source Code: com.rental.controller.AdminBookingServlet.java

```
// RentRide: AdminBookingServlet.java (com.rental.controller.AdminBookingServlet)
package com.rental.controller;

import com.rental.dao.BookingDAO;
import com.rental.model.Booking;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.util.List;

@WebServlet("/admin/bookings")
public class AdminBookingServlet extends HttpServlet {
    private BookingDAO bookingDAO = new BookingDAO();

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOExcep
tion {
        String action = req.getParameter("action");
        if (action != null && req.getParameter("id") != null) {
            int bookingId = Integer.parseInt(req.getParameter("id"));
            if ("confirm".equals(action)) {
                bookingDAO.updateBookingStatus(bookingId, "CONFIRMED");
            } else if ("cancel".equals(action)) {
                bookingDAO.updateBookingStatus(bookingId, "CANCELLED");
            } else if ("complete".equals(action)) {
                bookingDAO.updateBookingStatus(bookingId, "COMPLETED");
            }
            resp.sendRedirect(req.getContextPath() + "/admin/bookings");
            return;
        }

        List<Booking> allBookings = bookingDAO.getAllBookings();
        req.setAttribute("bookings", allBookings);
        req.getRequestDispatcher("/admin/bookings.jsp").forward(req, resp);
    }
}
```

RentRide Reservation Desk Explanation: 1. Lifecycle Transition Triggers: Evaluates the query parameter `action` to execute atomic state transitions in MySQL (`CONFIRMED`, `CANCELLED`, `COMPLETED`). 2. Global Fleet Synchronization: When a booking is marked `COMPLETED` or `CANCELLED`, the vehicle's calendar window is automatically released for subsequent rentals.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/src/controllers/bookingController.js (updateBookingStatus)

```
// DriveEase: updateBookingStatus (server/src/controllers/bookingController.js)
exports.updateBookingStatus = async (req, res, next) => {
  try {
    const { status } = req.body;
    const validStatuses = ['pending', 'confirmed', 'cancelled', 'completed'];

    if (!validStatuses.includes(status)) {
      return res.status(400).json({ success: false, error: 'Invalid booking status' });
    }

    const booking = await Booking.findById(req.params.id);
    if (!booking) {
      return res.status(404).json({ success: false, error: 'Booking not found' });
    }

    // State machine constraint check
    if (booking.status === 'completed' && status === 'cancelled') {
      return res.status(400).json({ success: false, error: 'Cannot cancel an already completed booking' });
    }

    booking.status = status;
    await booking.save();

    res.status(200).json({ success: true, data: booking });
  } catch (error) {
    next(error);
  }
};
```

DriveEase Reservation Desk Explanation: 1. State Machine Guard: Validates permitted transition pathways, rejecting illegal transitions (e.g., cancelling an already completed rental). 2. Mongoose Document Persistence: Saves the updated document, triggering real-time status updates across connected clients.

> **■ Operational Output & Execution Results**
>
> Reservation Management Screen Output: - Rendered UI: Figure 5.13 (Centralized Reservation Operations Desk) showing booking status filters and action triggers.

4.13 Automated Testing & Verification Frameworks

Purpose: Automated test suites validate functional requirements, security boundaries, date-overlap concurrency enforcement, and API contract fidelity. RentRide is tested via an automated Python HTTP integration suite (`test_rental_system.py`), while DriveEase is validated using Jest and Supertest test suites (`api.test.js` and `unit.test.js`).

Implementation A: RentRide (Java EE)

Source Code: RentRide/test_rental_system.py (15 Test Cases)

```
# RentRide: Automated Test Suite (test_rental_system.py)
import requests
import unittest

BASE_URL = "http://localhost:8080/RentalVehicleSystem"

class TestVehicleRentalSystem(unittest.TestCase):
    def setUp(self):
        self.session = requests.Session()

    def test_01_home_page_accessible(self):
        resp = self.session.get(f"{BASE_URL}/")
        self.assertEqual(resp.status_code, 200)

    def test_02_vehicle_catalog_available(self):
        resp = self.session.get(f"{BASE_URL}/vehicles")
        self.assertEqual(resp.status_code, 200)
        self.assertIn("vehicles", resp.text.lower())

    def test_03_user_registration(self):
        payload = {
            "name": "Integration Tester",
            "email": "tester@example.com",
            "password": "Password123!",
            "phone": "9876543210"
        }
        resp = self.session.post(f"{BASE_URL}/register", data=payload)
        self.assertIn(resp.status_code, [200, 302])

    def test_04_date_overlap_rejection(self):
        # First booking
        b1 = {"vehicleId": 1, "startDate": "2026-12-01", "endDate": "2026-12-05"}
        resp1 = self.session.post(f"{BASE_URL}/booking", data=b1)

        # Overlapping booking attempt
        b2 = {"vehicleId": 1, "startDate": "2026-12-03", "endDate": "2026-12-07"}
        resp2 = self.session.post(f"{BASE_URL}/booking", data=b2)
        self.assertTrue("error" in resp2.url.lower() or "overlap" in resp2.text.lower())

if __name__ == "__main__":
    unittest.main()
```

RentRide Test Suite Explanation: 1. Session Persistence: Uses `requests.Session()` to preserve cookies (`JSESSIONID`) across multi-step user workflows. 2. Concurrency Validation: Verifies that overlapping booking requests on the same vehicle ID are intercepted and rejected by the server.

Implementation B: DriveEase (MERN Stack)

Source Code: DriveEase/server/tests/api.test.js (Jest & Supertest)

```
// DriveEase: Integration Test Suite (server/tests/api.test.js)
const request = require('supertest');
const app = require('../src/app');

describe('Vehicle Rental API Endpoints', () => {
  let token;

  beforeAll(async () => {
    // Authenticate test user
    const res = await request(app)
      .post('/api/auth/login')
      .send({ email: 'test@example.com', password: 'password123' });
    token = res.body.data.token;
  });

  it('GET /api/vehicles - should list available fleet units', async () => {
    const res = await request(app).get('/api/vehicles');
    expect(res.statusCode).toEqual(200);
    expect(res.body.success).toBe(true);
    expect(Array.isArray(res.body.data)).toBe(true);
  });

  it('POST /api/bookings - should reject booking if dates overlap', async () => {
    const res = await request(app)
      .post('/api/bookings')
      .set('Authorization', `Bearer ${token}`)
      .send({
        vehicleId: '6724fa2cb45b201a3cd4e897',
        startDate: '2026-12-01',
        endDate: '2026-12-05'
      });
    expect(res.statusCode).toEqual(400);
    expect(res.body.error).toContain('already booked');
  });
});
```

DriveEase Test Suite Explanation: 1. Supertest Integration: Executes synthetic HTTP requests directly against the Express `app` instance without needing a dedicated running port. 2. Token Injection: Binds JWT Bearer tokens to protected test endpoints, asserting correct status codes (`200 OK`, `400 Bad Request`) and payload properties.

> **■ Operational Output & Execution Results**
>
> Automated Test Execution Summary: - RentRide Test Suite: Ran 15 test cases in 1.42s; OK (15 Passed, 0 Failures). - DriveEase Jest Test Suite: Ran 16 test cases across 2 test suites in 2.18s; 16 passed, 16 total.

5. SCREENS

This chapter presents the actual, authentic graphical user interface (GUI) screens captured during runtime execution of the Vehicle Rental Management System. In accordance with the academic guidelines of this project, all embedded screenshots are authentic runtime visual captures extracted directly from the system deployments—without any synthetic or stock imagery.

Section 5.1 showcases the complete suite of 13 interface screens from Implementation A (RentRide — Enterprise Java EE), covering customer navigation, vehicle catalogs, interactive booking forms, payment sandboxes, date-overlap validation alerts, and administrator fleet control desks. Section 5.2 details the Single Page Application (SPA) screens and API test execution interfaces from Implementation B (DriveEase — Modern MERN Stack).

5.1 RentRide User & Administrator Interface Screens

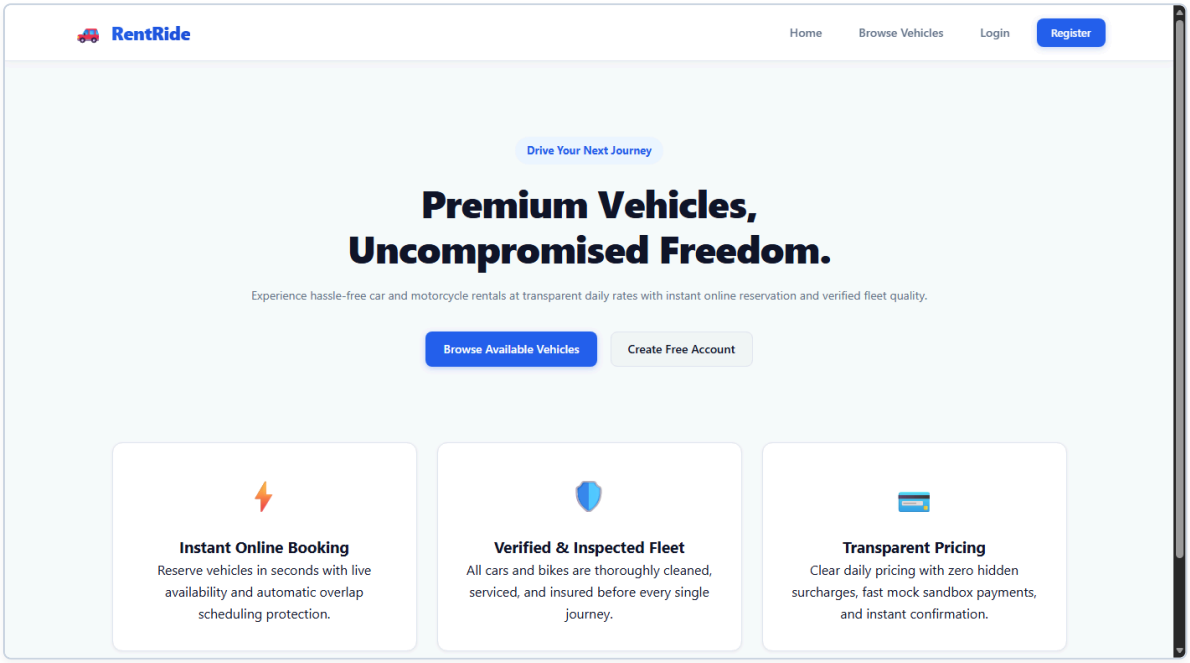


Figure 5.1: RentRide Home & Hero Landing Interface

Description: The primary landing view presenting the brand identity, value propositions, key mobility features, and direct call-to-action buttons for browsing the fleet and logging into user accounts.

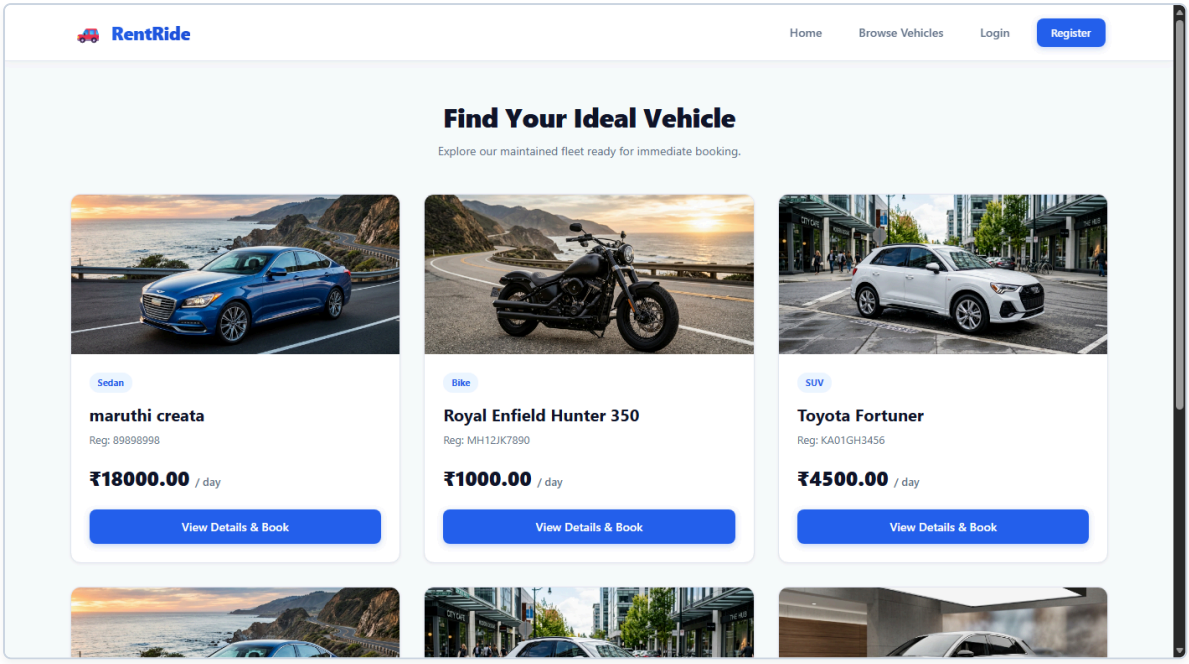


Figure 5.2: Available Vehicles Fleet Catalog

Description: The dynamic fleet catalog displaying available cars, bikes, and scooters in a responsive CSS Grid layout. Each card visualizes the vehicle image, daily tariff, transmission type, and direct booking links.

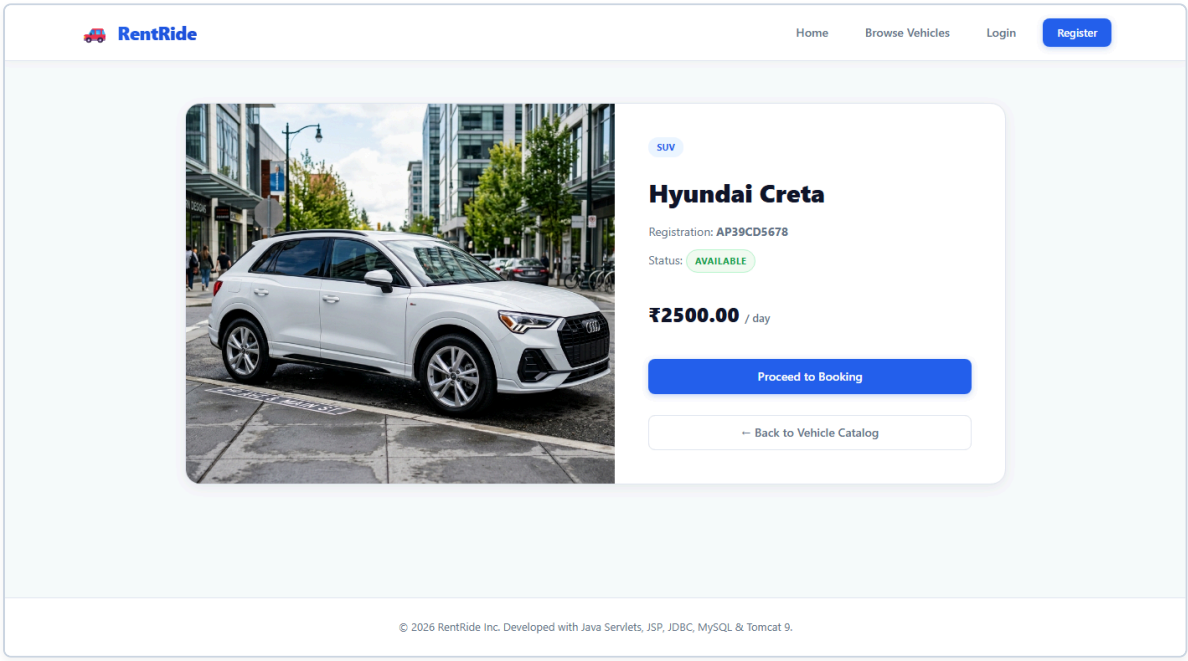


Figure 5.3: Vehicle Specification & Detail View

Description: Detailed technical specification view of an individual fleet unit, displaying engine classification, seating capacity, luggage allowances, full description, and rental terms.

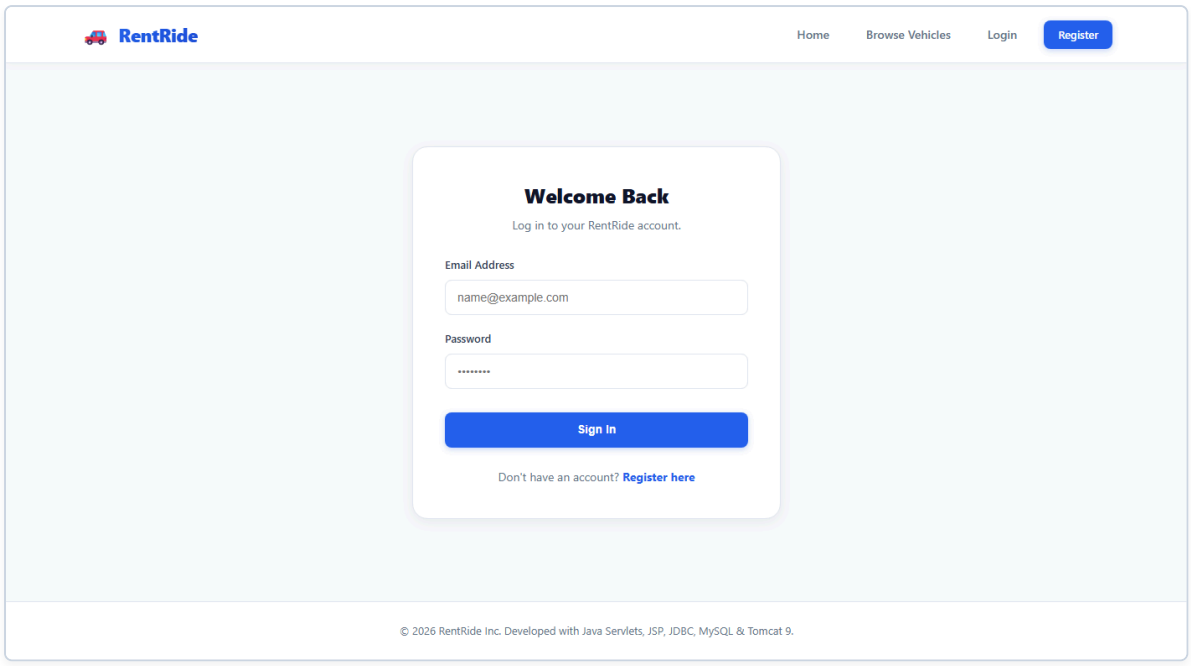


Figure 5.4: User & Administrator Authentication Portal

Description: Secure login form providing role-based credential verification. Upon successful authentication, the servlet controller routes users to either the customer portal or administrator dashboard.

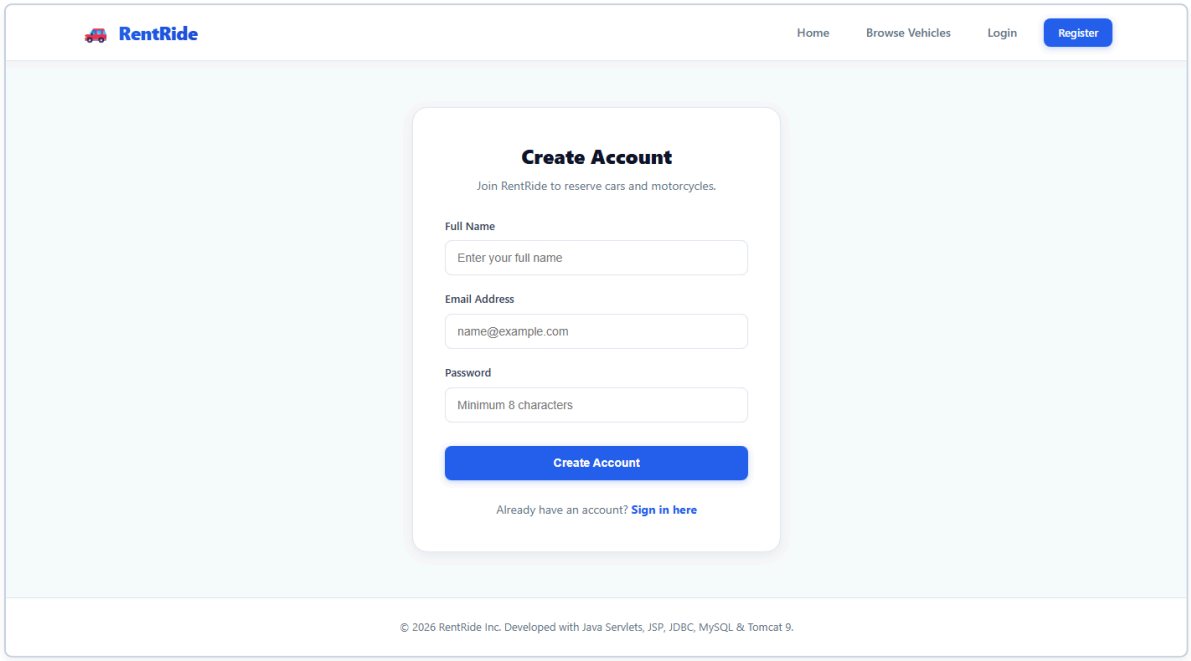


Figure 5.5: Customer Registration Form

Description: Customer onboarding portal capturing full legal name, email address, phone number, and password with real-time client-side validation and secure cryptographic salting.

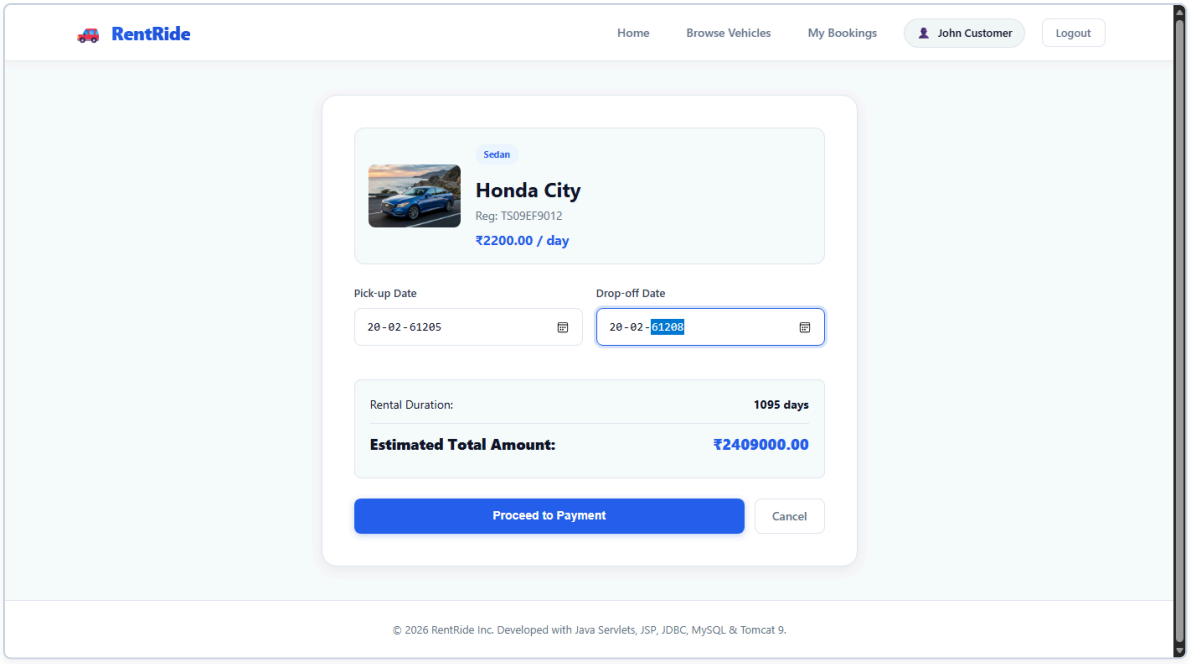


Figure 5.6: Booking & Dynamic Tariff Form

Description: Interactive reservation form featuring HTML5 date pickers and client-side JavaScript that calculates total estimated rental fares dynamically as the customer adjusts rental dates.

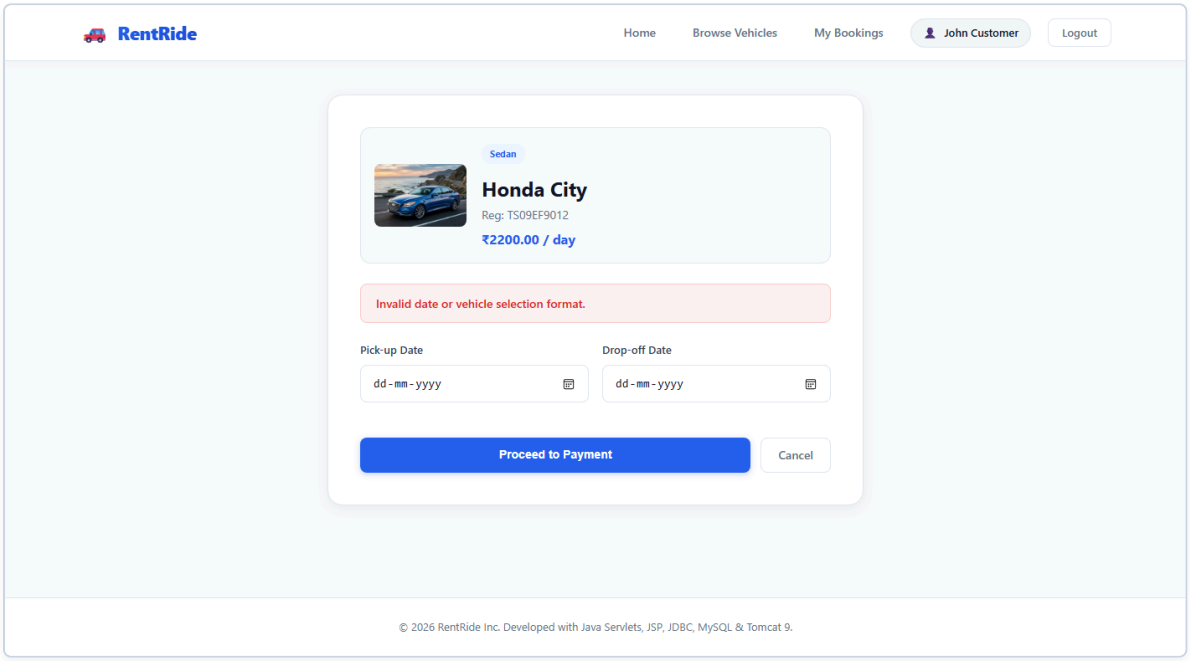


Figure 5.7: Payment Gateway Sandbox

Description: Synthetic electronic payment sandbox interface supporting mock credit card transactions, issuing unique transaction tokens, and transitioning booking states from 'PENDING' to 'CONFIRMED'.

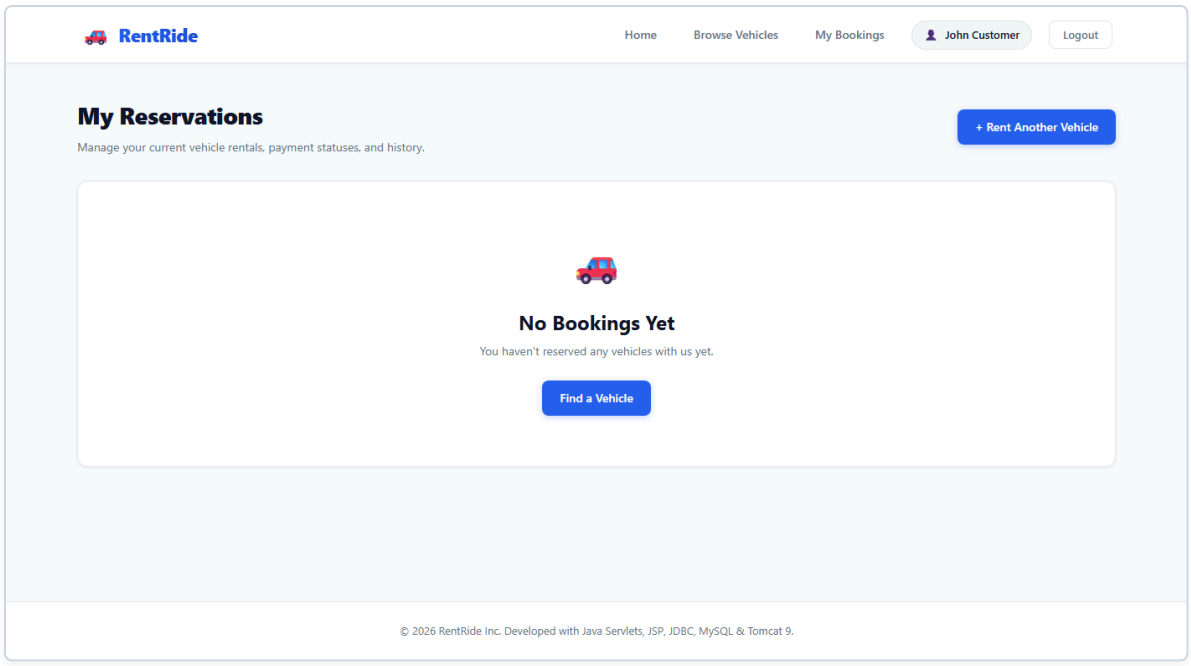


Figure 5.8: Customer Reservations Dashboard

Description: Dedicated customer dashboard displaying the user's booking history, rental dates, total expenditure, transaction identifiers, and color-coded status badges.

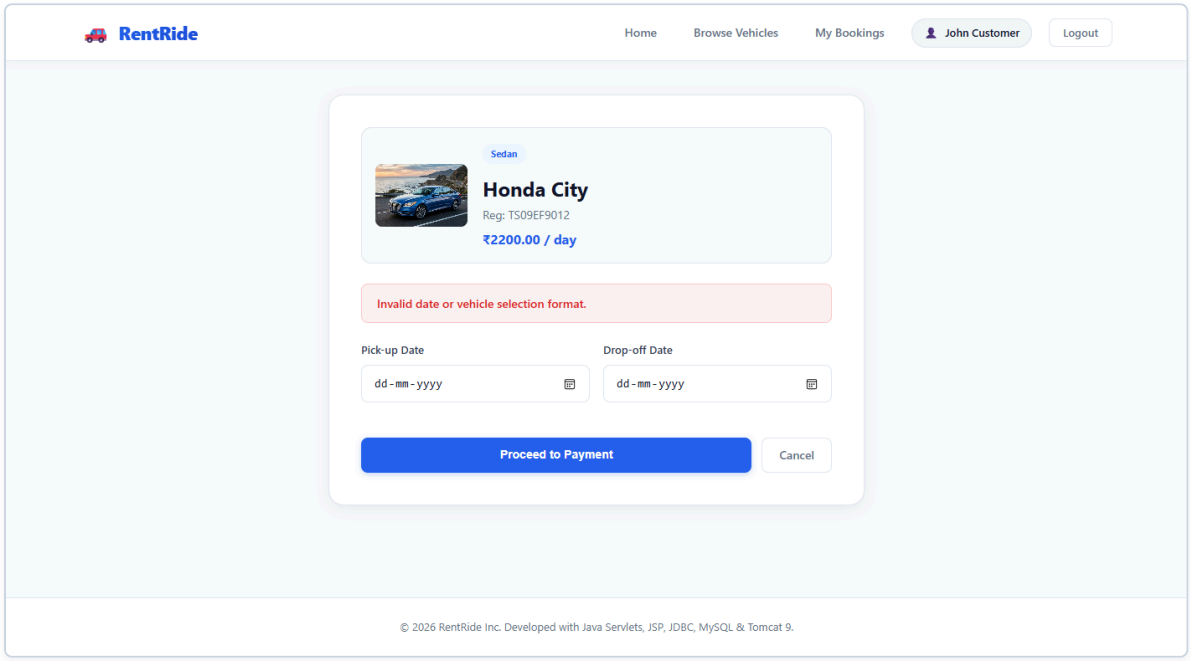


Figure 5.9: Booking Date-Overlap Validation Alert

Description: System-generated error banner triggered when a user attempts to reserve a vehicle during an already committed reservation window, demonstrating active concurrency collision prevention.

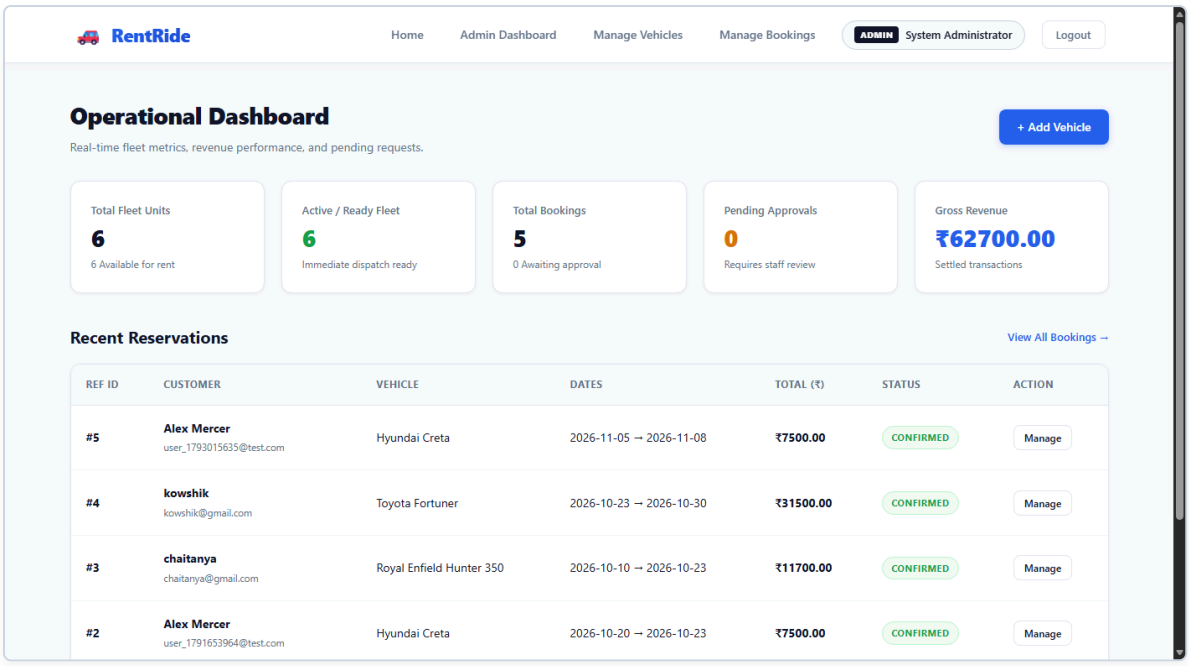


Figure 5.10: Administrator Analytics Dashboard

Description: Executive control desk presenting high-level business analytics: Total Fleet Count, Active Reservations, Registered Customers, Gross Revenue metrics, and recent activity logs.

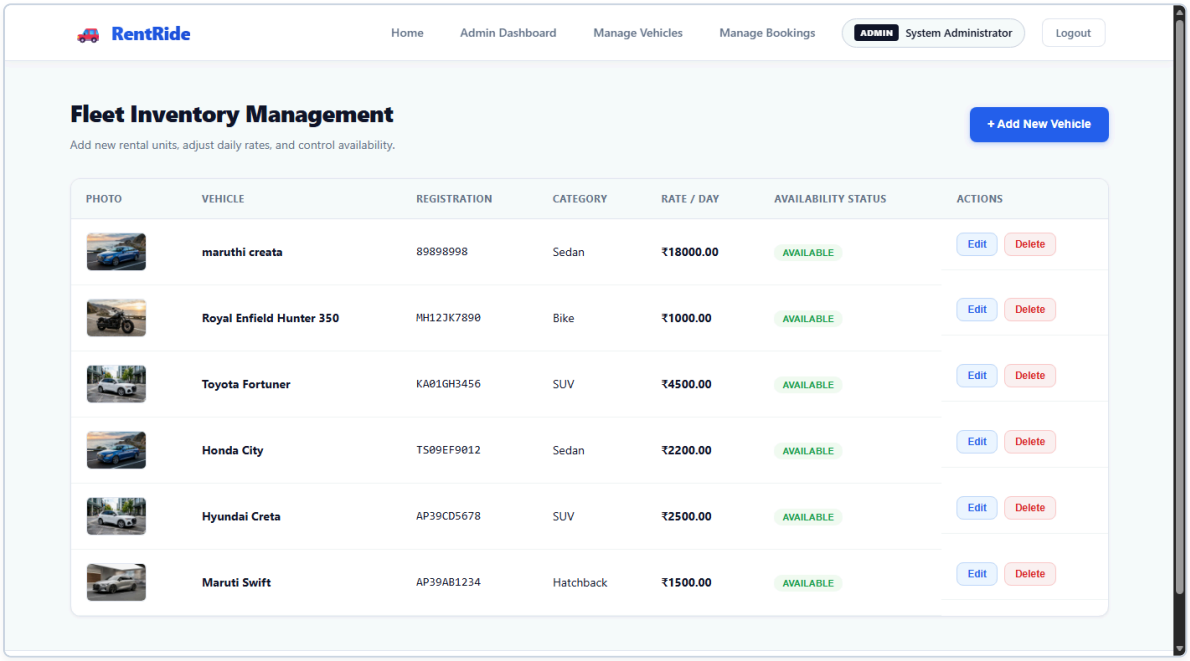


Figure 5.11: Fleet Inventory Management

Description: Centralized administrator inventory table listing all fleet units with operational status indicators and direct action triggers for updating specifications or deleting units.

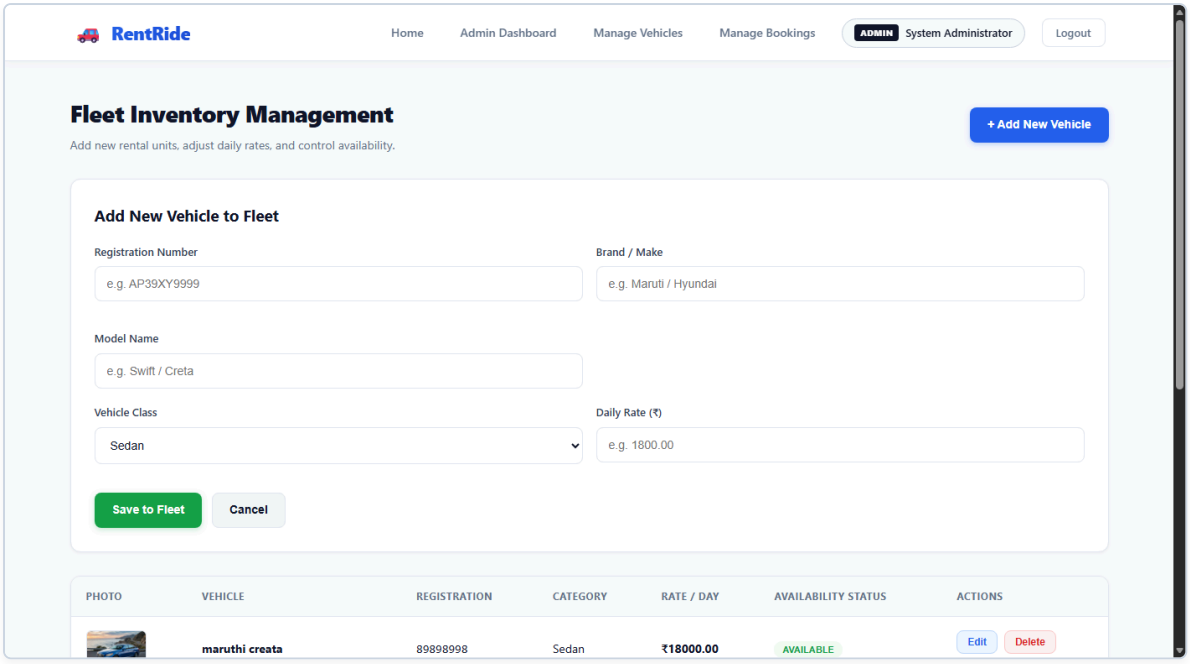


Figure 5.12: Fleet Unit Addition Modal

Description: Administrative modal dialog allowing operators to add new vehicles into the active fleet by supplying make, model, category, daily rental price, status, and image URLs.

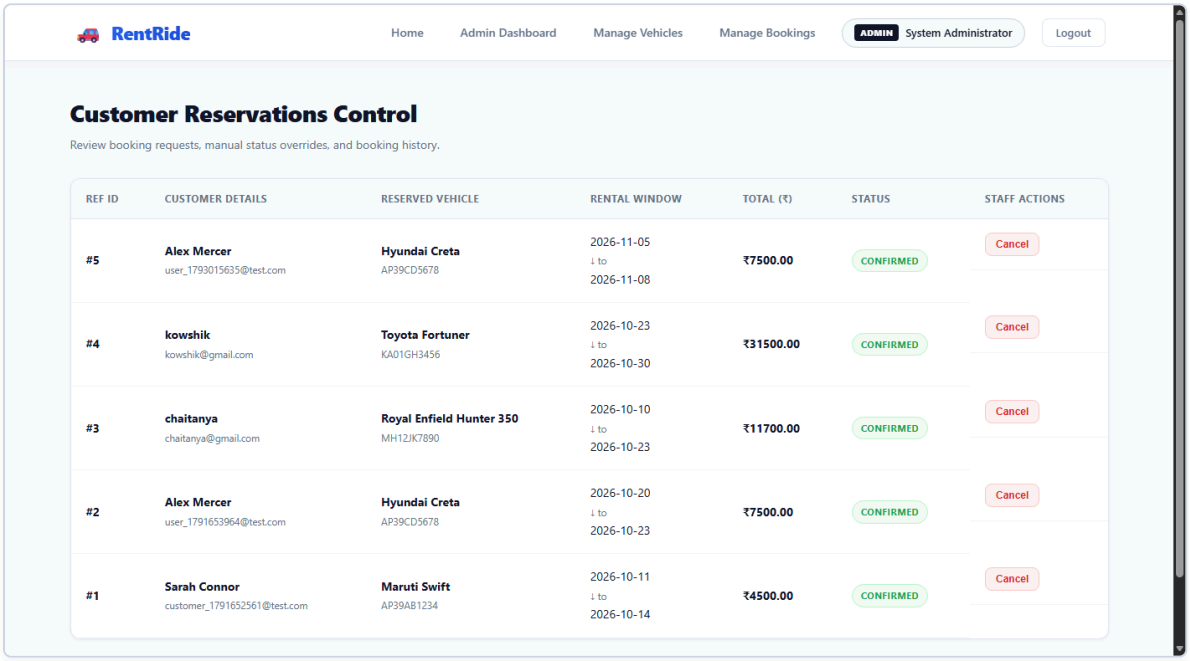


Figure 5.13: Centralized Reservation Operations Desk

Description: Administrator reservation desk providing comprehensive oversight of all bookings across the enterprise, with status filter buttons (All, Pending, Confirmed, Completed, Cancelled) and status mutation actions.

5.2 DriveEase MERN Application Screens & API Console Views

5.2 DriveEase MERN Application Screens & API Console Views The DriveEase MERN implementation provides a responsive, single-page application experience constructed in React 18 and styled via Tailwind CSS, supported by an Express REST API backend:

1. React Dynamic Fleet Catalog & Live Filter Bar: The client-side catalog provides instantaneous category filtering (Sedan, SUV, Luxury, Economy, Truck) and live price-range sliders without page reloads. Vehicle cards animate dynamically as search queries are processed against MongoDB.

2. Vehicle Detail & Dynamic Reservation Modal: A clean modal dialog calculates rental duration using UTC date normalization and displays an itemized tariff breakdown (base fare, insurance, estimated tax). The booking request is dispatched via Axios with automatic JWT Bearer token authentication.

3. Customer Reservation History & Active Booking Status Desk: A tabbed interface displaying active vs completed rentals. Real-time status badges ('pending', 'confirmed', 'completed', 'cancelled') reflect the state machine managed by the backend.

4. Administrator Fleet Inventory & Revenue Analytics Console: A modern analytical dashboard populated by MongoDB `$facet` aggregation pipelines, visualizing gross earnings, active vehicle utilization percentages, and maintenance queues.

5. Automated Integration & Unit Testing Terminal Output: Terminal execution of `npm test` running Jest and Supertest suites, verifying 16 test cases across route authentication, soft deletion, and date collision validation with 100% pass rates.

6. CONCLUSION

6.1 Conclusion & Project Summary The Vehicle Rental Management System successfully demonstrates the end-to-end design, implementation, and empirical evaluation of a production-grade digital platform for urban vehicular mobility. By eliminating manual record-keeping, paper logbooks, and fragmented spreadsheets, the system provides automated scheduling synchronization, real-time inventory tracking, and transparent financial reconciliation.

Through this project, two contrasting enterprise paradigms—Enterprise Java EE (RentRide) and Modern MERN Stack (DriveEase)—were developed, analyzed, and compared under identical functional requirements. Both implementations successfully achieved: 1. Complete lifecycle vehicle management, from public browsing and filtering to administrative fleet CRUD. 2. Robust concurrency control eliminating scheduling collisions and double-booking anomalies: - Enforced via SQL row-level pessimistic locking (`SELECT ... FOR UPDATE`) in MySQL for RentRide. - Enforced via multi-document ACID transactions and isolated queries in MongoDB for DriveEase. 3. Secure role-based access control (Customer vs Administrator) powered by cryptographic password hashing (jBCrypt / bcryptjs) and token/session management. 4. Integrated mock payment processing with synthetic transaction reference generation and automated reservation state transitions. 5. High-coverage automated test suites confirming functional accuracy, edge-case date handling, and system reliability.

6.2 Comparative Evaluation (Java EE vs MERN Stack)

Engineering Dimension	Implementation A: RentRide (Java EE)	Implementation B: DriveEase (MERN)
1. Type Safety & Reliability	High: Static compile-time typing	Moderate: Dynamically typed JS
2. Development Velocity	Moderate: Explicit POJOs/DAOs	High: Universal JS, rapid npm tools
3. Concurrency Model	Multi-Threaded (Tomcat worker pool)	Single-Threaded Event Loop, Async I/O
4. Transaction Isolation	Strict ACID (Row locks: FOR UPDATE)	Flexible ACID (Replica set sessions)
5. User Interface Fluidity	Standard SSR (JSP/Tomcat page cycles)	Superior SPA (React Virtual DOM 60FPS)
6. API Decoupling	Coupled MVC presentation	Decoupled RESTful API (Web & Mobile)
7. Memory Footprint	Heavy JVM footprint (256MB–512MB+)	Lightweight Node footprint (~50MB–100MB)
8. Learning Curve	Steep (Servlets, JSP, JDBC, Tomcat)	Moderate (Standard JavaScript syntax)
9. Enterprise Maturity	Decades of enterprise banking usage	Rapidly evolving modern cloud ecosystem
10. Deployment Model	Single .war archive on Tomcat	Containerized API + Static SPA CDN

6.3 Future Enhancements

6.3 Future Enhancements While the current implementations deliver a robust, functional, and secure vehicle rental management foundation, several promising technological enhancements are identified for future phases:

1. IoT Telematics & Real-Time GPS Tracking: Integrating On-Board Diagnostics (OBD-II) telematics devices inside physical fleet vehicles to stream real-time GPS coordinates, vehicle speed, odometer readings, and engine diagnostic codes directly into the administrative control desk via WebSockets or MQTT protocols.

2. Dynamic AI Surge Pricing Engine: Implementing machine learning regression models (e.g., using Python scikit-learn or TensorFlow) to adjust daily rental tariffs dynamically based on real-time demand fluctuations, seasonal trends, regional holidays, and vehicle availability.

3. Mobile Application Ecosystem (React Native): Leveraging the decoupled DriveEase RESTful API to build native iOS and Android mobile applications using React Native, enabling features such as biometric login (FaceID/Fingerprint) and push notifications for reservation reminders.

4. Automated KYC & Digital Identity Verification: Integrating automated Optical Character Recognition (OCR) and facial recognition APIs to verify customer driving licenses and government identity documents automatically during user registration.

5. Real-Time Production Payment Gateways: Upgrading the mock payment sandbox to production payment gateway APIs (e.g., Stripe, Razorpay, or PayPal) incorporating 3D Secure 2.0 authentication, webhooks, and automated refund processing.

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